

Intrinsic-Dimension-Guided Bottleneck Selection Revisited: Search Policies, Quality Targets and Diagnostic Limits

Chika Obata ¹ , Mizuki Dai ¹ , and Kenya Jin'no ^{1,*} 

¹ Department of Intelligent Systems, Faculty of Information Engineering, Tokyo City University, 1-28-1 Tamazutsumi, Setagaya-ku, Tokyo 158-8557, Japan; g2332016@tcu.ac.jp (C.O.); g2591402@tcu.ac.jp (M.D.)

* Correspondence: kjinno@tcu.ac.jp; Tel.: +81-3-5707-0104 (K.J.)

Abstract

Latent-width selection depends on both diagnostic information and the search policy using it. We retrospectively compare six policies on stored variational-autoencoder candidates from four image datasets, five training seeds and five regularisation weights. A validation anchor defines the quality target, and one canonical timing snapshot prices every request. The historical intrinsic-dimension window incurs structurally redundant work, so its expense contrast alone cannot identify the value of dimension information. A revised local policy and a matched midpoint control expose the trade-off between requests and the smallest qualifying width. At the inherited defaults, the dimension-initialised policy matches the full-grid minimum in all 100 conditions, while the midpoint control matches in 98; the former uses ascending fallback on three datasets. Across 10,000 retrospective coefficient/threshold settings, 155 local outputs miss the grid minimum despite meeting quality. Reference-bank acquisition and reuse change the cost interpretation. A new one-seed MNIST experiment evaluates Monte Carlo ELBO selection, while pruning adaptations remain separate implementation audits. Active-unit prediction does not establish an additional benefit on its near-ceiling task. These results support explicit search, quality and cost specifications rather than a universal ranking of dimension-selection methods.

Keywords: variational autoencoder; latent dimension; intrinsic dimension; active units; model selection; reconstruction quality; reproducibility

1. Introduction

The latent width of a variational autoencoder (VAE) [1] controls the capacity available to its representation. Choosing that width by validation over a candidate grid is straightforward, but requires training several models. Alternatives include intrinsic-dimension (ID) estimation [2,3], comparisons between sampled and mean latent representations [4], activity diagnostics [5,6], and pruning through reconstruction constraints or relevance priors [7,8]. These signals answer different questions: geometric dimension, coordinate use, reconstruction quality and network width need not coincide.

This study asks how the use of ID changes the decisions and training requests of a latent-width search, whether AU diagnostics add predictive value beyond other recorded quantities, and how these measurements depend on quality targets and readout settings. We compare policies on common candidate-model families and fixed splits. Search order, stopping rules, quality attainment and agreement with the smallest qualifying grid width are measured separately.

Received:

Revised:

Accepted:

Published:

Copyright: © 2026 by the authors.

Submitted to *Mach. Learn. Knowl. Extr.* for possible open access publication under the terms and conditions of the

[Creative Commons Attribution \(CC BY\) license](https://creativecommons.org/licenses/by/4.0/).

The contribution is a retrospective comparison of search policies and their accuracy–cost trade-off, together with a record-level audit. We contrast a historical fixed-window policy with an ID-initialised local search, an otherwise identical search without ID, ascending search and a full-grid reference. Four image datasets, five candidate seeds and five regularisation weights are covered. We report the full factorial sensitivity to the ID coefficient and reliability thresholds, and separate the cost of acquiring an ID bank from its reuse.

The inherited pruning implementations are treated as implementation audits, outside the main policy comparison. A normalized relevance readout and a new, bounded native-model diagnostic test specific implementation issues; neither is presented as a reproduction of an original pruning method. A separate MNIST retraining experiment evaluates Monte Carlo ELBO capacity selection. Sections 3–5 define the protocols, Section 6 reports results, and Section 7 states the scope of inference.

2. Related Work

2.1. Dimension, Reconstruction and Latent Activity

PCA [9], Isomap [10] and locally linear embedding [11] construct low-dimensional representations. ID estimators instead quantify local dimension from sample geometry. We use TwoNN and an MLE cross-check; other approaches include DANCo, expected simplex skewness and geodesic entropic graphs [12–14]. Estimator limitations and dependence on the sampled distribution motivate cross-checks [15,16]. Graph-based methods are outside this experiment’s scope; we do not infer that they would resolve or reproduce the observed disagreements. Minimal-spanning-tree isolation forests also use graph geometry for anomaly scoring [17]; that objective is distinct from selecting a VAE width under a reconstruction target. Lower ID in learned representations has also been observed in deep networks [18,19].

FONDUE [4] searches for the capacity preceding a separation between ID estimates of sampled and posterior-mean representations. Its appendix gives FONDUE-VAR, which chooses a count from latent-variable types. AU and variable types share an interest in unused coordinates but are different statistics: AU thresholds the across-data variance of posterior means, whereas the type classifier uses moments of posterior variances [6]. Agreement of their final counts is not equivalence of their stability or information content.

GECO with L_0 regularisation optimises a gated VAE under a reconstruction constraint, with KL and gate-count terms [7,20]. The source uses ARM; our adaptation uses hard-concrete gates [21]. Neither the nonconvex optimisation nor our implementation guarantees a global minimum capacity. Our validation anchor plus tolerance is a particular way of setting an external quality yardstick, not an identical implementation of the source’s constraint.

ARD-VAE uses a hierarchical relevance prior [8]. The source tunes a regularisation weight, uses a Gaussian approximation for its closed-form KL, and selects axes explaining 99% of a Jacobian-weighted variance score. It is therefore not threshold-free. The local adaptation has additional differences, including a finite-step decoder score and a fixed selection-stage weight; these are detailed in Appendix C.2.

2.2. Relation to the Conference Paper and Earlier Manuscript

Our NOLTA2025 contribution [22] concerned the relation between input ID and latent-space capacity. The available conference manuscript describes class-specific MNIST experiments with four fully connected and two convolutional architectures and four seeds. CNNs and multiple seeds are consequently not new features of this revision. The conference manuscript available for comparison has a working title different from the published title;

the reference uses the title and pages listed by the authors’ institutional publication record. Table 1 distinguishes this earlier scope from the present comparison. MAKE47 denotes the earlier journal submission, not a separate publication.

Table 1. Scope of the conference study, earlier journal submission and present revision.

Item	NOLTA2025	MAKE47	Present revision
Question	Input and latent ID versus capacity	ID-guided choice with AU verification	Selection quality, cost and diagnostic limitations
Models	Four FC and two CNN structures; MNIST classes; four seeds	FC/CNN and synthetic/image extensions	Fixed candidate families on four image datasets; five candidate seeds
Comparators	Architecture and capacity comparisons	Capacity sweeps and diagnostic checks	FONDUE variants, pruning adaptations, grid criteria and ascending search
Readouts	ID estimates	AU-based operational checks	Same-checkpoint type/AU trajectories and paired predictive analysis
Evidence	Conference settings	Earlier loss convention	Corrected-loss records; explicit targets, final test metrics and accounting limits

3. Evaluation Framework

3.1. Models and Quality Target

We distinguish the *ID-reference AE*, which supplies a representation for estimation, the *quality anchor*, an ordinary VAE at the largest candidate width, and the *selected model*, the ordinary VAE evaluated after dimension selection. Write $\mathcal{M}$ for the candidate grid, s for a training seed, and $D_s(m, \beta)$ for validation per-pixel MSE of the deterministic reconstruction $g(\mu(x))$. The implemented target is

$$m_a = \max \mathcal{M}, \quad T_{D,s}(\beta) = D_s(m_a, \beta) + \max\{0.10D_s(m_a, \beta), 0.001 \operatorname{Var}(X_{\text{val}})\}. \quad (1)$$

The variance is computed over all validation-array entries. The relative term dominates in every recorded image condition, giving $T_{D,s} = 1.10D_s(m_a, \beta)$. The configuration originally also specified a three-seed-SD term, but the executed target function was supplied one anchor at a time and omitted that term. We report the implemented rule; we do not retrospectively describe it as a pooled-seed target. Appendix A lists every dataset- β anchor and target.

For a fixed seed and β , the target is held constant during search. It is recomputed from a different anchor when β changes. A β -sweep therefore changes both the learned representation and the quality yardstick. It is not a fixed-quality intervention. A method that returns a valid width need not meet this external target; quality attainment is evaluated separately for every final model.

3.2. Canonical Replay Prices and Historical Ledgers

Replay uses one immutable candidate snapshot. A logical key contains dataset, width, training seed, requested epochs, early-stopping option and β ; architecture, likelihood and split are fixed by the dataset configuration. Each unique requested key, including the quality anchor, is charged its duration in this snapshot once. Metrics and prices are therefore identical whenever replay policies request the same key. These are paired uses of stored records, not independent retraining.

Earlier method ledgers sometimes attach different durations to the same logical key. Even when validation and test summaries agree exactly, missing historical weights and execution identifiers prevent proving checkpoint identity. We therefore do not mix those prices in the new policy comparison. Appendix B.1 preserves the historical account and resolves every anchor/final role to its key, source and both available durations in a supplementary CSV. The quality anchor and its price in each replay come from the same record.

Let C_{cand} sum the canonical prices of requested candidates and C_{ref} be the nine-AE bank's recorded training cost. Cold use charges $C_{\text{cand}} + C_{\text{ref}}$ for an ID policy; warm use charges C_{cand} when that bank already exists. A 25-task batch (five seeds by five weights for one dataset) allocates $C_{\text{ref}}/25$ per task. This does not turn shared-bank outcomes into independent ID replications. All accounts exclude missing ID-computation and some post-training evaluation times. They estimate attributed training costs, not new end-to-end wall-clock measurements. Historical probe and FONDUE calibration costs are reported separately from the replay policies.

3.3. Outputs and Termination

Each trial records a raw returned dimension, the width of the evaluated model, the ID reliability decision where applicable, fallback use, the logged termination state, the budget flag, quality attainment, and whether the final width is smaller than the anchor. These events can overlap. In particular, fallback can produce a compact model that meets the target. The label SUCCESS in an older log indicates ordinary termination; it is not itself evidence of target attainment.

A budget-exceeded trial contributes its consumed time and, if it has a returned model, that model's dimension and metrics. A raw dimension below one is reported as degenerate; its time and failure count remain, while MSE is missing because no final model exists. The original implementation checks budgets after completed calls and can still obtain a final model. The reported limits are therefore soft accounting limits, not hard interruptions. Outcome tables preserve these distinctions.

3.4. Reference Bank and Reliability Checks

The reference bank uses three widths and three fixed reference seeds, shared across the five candidate seeds. Let $\bar{d}_r$ be the mean TwoNN estimate at reference width r , $\hat{d} = \text{mean}_r \bar{d}_r$, and $\bar{d}_{\text{MLE}}$ the mean MLE estimate over that same bank. The checks are

$$R = \frac{\max_r \bar{d}_r - \min_r \bar{d}_r}{\text{median}_r \bar{d}_r} \leq 0.15, \quad A = \frac{|\hat{d} - \bar{d}_{\text{MLE}}|}{\hat{d}} \leq 0.25. \quad (2)$$

The values $c = 2$, $R_{\text{max}} = 0.15$ and $A_{\text{max}} = 0.25$ were internal heuristics recorded in the 12–13 September specification. The factor two was geometrically motivated but not derived for this statistical task; neither reliability tolerance was calibrated to a false-rejection rate. Their numerical roles are examined retrospectively rather than justified by the quality-margin experiment.

3.5. Retrospective Search Policies

We define the following replay on 19 September, after the earlier results and review. A full-grid reference returns $m^* = \min\{m \in \mathcal{M} : D_s(m, \beta) \leq T_{D,s}\}$. Ascending search obtains the anchor and visits increasing widths until the first pass; it identifies m^* without a monotonic-error assumption. The historical ID procedure exhausts a lower window when ID is accepted and uses minimum-MSE coarse search otherwise. Its full executed pseudocode is retained in Appendix C.1.

The revised *ID-local* policy uses ID to choose a starting width rather than an exhaustive window. If the start passes, it walks downwards to the first failure; otherwise it walks upwards to the first pass. This can omit many candidates, but can miss a smaller qualifying width if the pass/fail sequence is not monotone. Rejected ID banks now invoke ascending- Q fallback, not the historical coarse-MSE fallback. This fallback change is part of the retrospective policy.

Algorithm 1 (ID-local replay and its matched controls).

Input: grid, seed, weight, validation records and canonical prices, reference bank, c , $R_{\max}$, $A_{\max}$; no test metrics.

1. Request the largest width once; compute its target from Equation 1. A pass means validation MSE at most that target.
2. For ID-local, charge the reference bank and compute Equation 2. If either threshold fails, request ascending widths and return the first pass.
3. Otherwise start at the smallest grid width at least $c\hat{d}$, clipped to the largest width. For the midpoint control, omit all reference costs and instead start at zero-based index $\lfloor (|\mathcal{M}| - 1)/2 \rfloor$.
4. If the start fails, request larger widths in order and return the first pass. The anchor guarantees a returned qualifying width.
5. If the start passes, request adjacent smaller widths in descending order until one fails, or the smallest width is reached; return the last pass.
6. For the optional certified ID variant, test previously unqueried widths below the local answer in ascending order, stopping at the first pass. This certifies the full-grid minimum without monotonicity.
7. Charge each distinct candidate once. Return width, requested keys, count, reference/candidate costs and the soft-budget flag. Attach test metrics only after the selection is fixed.

Full-grid- Q , ascending- Q , the legacy path, ID-local, midpoint-local and certified ID-local are replayed on all 100 dataset-weight-seed conditions. Paths are completed within the stored grid (at most 12 distinct candidates); the original time limit is recorded as a flag rather than interrupting this counterfactual evaluation. Legacy rows retain their executed path and output, including original fallback. We report quality, the width gap $m - m^*$, exact agreement, candidate requests and canonical cost. Midpoint-local controls the local stopping rule while removing ID acquisition; it is a simple control fixed for the replay, not a tuned or universally optimal starting point.

The coefficient set is $\{1, 1.5, 2, 3, 4\}$, inherited from the earlier configuration. The retrospective reliability grids are $R_{\max} \in \{0.10, 0.15, 0.25, 0.50\}$ and $A_{\max} \in \{0.15, 0.25, 0.40, 0.85, 2.00\}$. They include stricter settings and settings on either side of the observed dataset decisions. The most permissive values are stress tests, not recommended defaults. All 10,000 combinations of these parameters with the 100 recorded conditions are retained; no best-performing threshold replaces the original default.

4. Theoretical Background

For an ideal smooth d -manifold, Whitney's embedding theorem gives an embedding into $\mathbb{R}^{2d}$ [23]. Exact continuous reconstruction through an injective encoder entails an embedding, but neither a noisy finite-sample ID estimate nor an approximate neural reconstruction satisfies these assumptions automatically. The multiplier two is therefore a heuristic under evaluation, not a quality guarantee or a lower bound inferred from measured ID.

In linear Gaussian latent-variable models, informative directions depend on the covariance spectrum and noise scale; analyses of linear VAEs and posterior collapse make this relation explicit [9,24]. VAE alignment with principal directions has also been studied [25]. These results motivate measuring how regularisation affects coordinate use. They do not

equate AU with intrinsic or minimum embedding dimension in a nonlinear VAE. AU also depends on the latent coordinates and the threshold used to count them.

5. Experimental Setup

5.1. Data, Architecture and Optimisation

Table 2 gives the actual image-data settings. MNIST [26], Fashion-MNIST [27] and CIFAR-10 [28] use random subsets of the official training and test pools. A permutation with seed 20260912 selects 20,000 training-pool examples; another with seed 20260913 divides them 90:10. The 3000 test examples use seed 20260914. For dSprites [29], which has no official train/test split, the first 20,000 and next 3000 entries of one permutation of 737,280 images are disjoint pools; the 20,000 are divided 90:10 and each index list is sorted before loading. Images are scaled to $[0, 1]$, with no augmentation. Training seeds are 42, 123, 777, 2024, 31337; the split does not vary with these seeds.

Table 2. Image-data settings used for candidate VAEs. All datasets use 18,000 training, 2000 validation and 3000 test examples. G: Gaussian; B: Bernoulli. Budgets are recorded-time limits with a common limit of 12 candidate-training calls; ID AEs are additional.

Dataset	Input	Likelihood	Candidate widths	Budget (s)
MNIST	$1 \times 28 \times 28$	G	2, 4, 6, 8, 10, 12, 16, 20, 24, 32, 48, 64	1470
Fashion-MNIST	$1 \times 28 \times 28$	G	Same as MNIST	1470
dSprites	$1 \times 64 \times 64$	B	Same as MNIST	5163
CIFAR-10	$3 \times 32 \times 32$	G	8, 16, 24, 32, 48, 64, 96, 128, 192, 256	2924

For MNIST and Fashion-MNIST, the encoder has dense widths 512 and 256, ReLU activations, and separate linear mean/log-variance heads; the decoder has widths 256 and 512 with a linear output. The CIFAR-10 encoder has channels 32, 64 and 128; dSprites adds 256. Convolutions use kernel 4, stride 2, padding 1 and ReLU, leaving 4×4 feature maps, followed by linear posterior heads. A linear/ReLU projection and reversed transposed convolutions form the decoder, ending in a sigmoid. The ID-reference AE flattens the input and uses dense widths 512–256– r and the reverse decoder, with a linear output; it is separate from the candidate CNNs.

Ordinary candidates use Adam, learning rate 10^{-3} , default moments $(0.9, 0.999)$, $\epsilon = 10^{-8}$, no weight decay, batch size 128 and up to 300 epochs. The last batch has 80 examples (141 updates per epoch). ID AEs use the same optimiser and batch size for 100 epochs. Early stopping uses patience 30. A recorded plateau flag requires $(\max h - \min h) / |\text{mean } h| < 0.01$ over the last 50 validation-monitor values. It is not an additional stopping rule. Representative trajectory runs always complete 300 epochs.

The candidate training loss is

$$\mathcal{L}_\beta = \frac{1}{B} \sum_{i=1}^B \left[\ell(x_i, g(z_i)) + \beta \sum_{j=1}^m \frac{\mu_{ij}^2 + \exp(v_{ij}) - 1 - v_{ij}}{2} \right], \quad z_i \sim q_\phi(z | x_i), \quad (3)$$

where v is log variance. Gaussian reconstruction uses a sum of squared errors, equivalent to fixed observation variance $1/2$ up to an additive constant; dSprites uses summed binary cross-entropy. The primary weight is one; the ladder is $\{0.25, 0.5, 1, 2, 4\}$, constant during each run. No cyclical annealing schedule [30] is used. Equal numerical β across likelihoods does not imply equal regularisation strength.

The stored validation monitor named “ELBO” is $h = -[\ell(x, g(\mu(x))) + \text{KL}]$, averaged over examples. It uses posterior-mean reconstruction, not an expectation over posterior samples, and is therefore an *ELBO proxy*, not an unbiased ELBO estimate or a certified bound. It also uses unit KL weight for checkpoint selection even when training uses

another β . We retain these checkpoints and label the monitor accurately. No test metric enters the selection code. However, the candidate cache evaluates test metrics for every trained candidate, before method selection; the claim that only one final test evaluation was executed would be incorrect. Tables report test metrics only for the validation-selected models. This is a code-level separation of selection and evaluation, not a prospectively sealed test workflow.

5.2. Methods and Readouts

Grid search selects by validation MSE, the ELBO proxy, or linear-probe validation accuracy. Probes use posterior means, multinomial logistic regression, up to 2000 iterations, and $C \in \{0.01, 0.1, 1, 10\}$ chosen by validation. Labels are digit/apparel/ CIFAR classes and the three dSprites shapes. Both capacity and probe regularisation use validation labels in the probe selector. No centring, scaling or whitening is applied to the probe latents; logistic regression uses its default intercept. Coarse search uses Algorithm A1's subset; the fixed baseline uses width 32.

FONDUE is a local PyTorch implementation using TwoNN and a fixed ID-gap threshold $0.2d_0$. It searches integer widths, which can lie outside the ordinary candidate grid. Its selected width is trained exactly. At $\beta = 1$, short training uses one epoch for MNIST and dSprites and two for Fashion-MNIST and CIFAR-10. dSprites follows the source's epoch choice; elsewhere a seed-42 calibration increases epochs until two consecutive predictions agree, stopping at five. Equality is first tested after runs at one and two epochs; if predictions at $e - 1$ and e agree, the returned epoch count is $e - 1$, and both calibration costs remain charged. Other β -specific choices are in Appendix C. FONDUE+Q maps the returned width upwards onto the grid, obtains the anchor, and extends upwards until quality is met or its budget flag is set. This adds an external reconstruction rule to FONDUE.

FONDUE-VAR starts at round $(2d_0)$ and doubles width until the declared type condition is met. With mixed variables kept it returns active plus mixed once a passive variable appears; otherwise it returns the active count once a mixed or passive variable appears. We implement the description-based classifier: posterior variance is exponentiated *once* from log variance. A coordinate is passive if its across-data variance is below 0.1 and its mean is in $[0.9, 1.1]$, active if that variance is below 0.1 and its mean is at most 0.1, and mixed otherwise. The audited public-code version is identified in Appendix C.2; this is not claimed to reproduce every path of that code or its published experimental settings.

AU is independently defined by

$$AU_\delta = \sum_{j=1}^m \mathbf{1}\{\text{Var}_{x \in X_{\text{est}}} \mu_j(x) > \delta\}, \quad \delta \in \{10^{-3}, 10^{-2}, 10^{-1}\}, \quad (4)$$

with primary $\delta = 10^{-2}$. The across-example variances for AU, variable types and the target floor use denominator n (population variance); reported between-seed SDs use denominator $n - 1$. ID and activity readouts use the same 10,000 training examples sampled without replacement using seed 20260915. The MLE cross-check uses $k = 10$ neighbours. AU is diagnostic, not an extra selector in the method comparison.

The GECCO and ARD adaptations start at the largest grid width and run 300 epochs. Their raw coordinate counts are mapped to the nearest candidate width, with ties going to the smaller width, for the final ordinary VAE. Appendix C.2 specifies the local objectives, updates and readout differences. The evaluated MSE thus belongs to the transferred ordinary VAE, not to the original model.

5.3. Provenance and Scope of Prior Specification

The saved image-experiment provenance is Python 3.12.3, PyTorch 2.7.1+cu118, CUDA 11.8, cuDNN 90100 (9.1.0), NVIDIA driver 570.211.01, and an NVIDIA GeForce RTX 3070, on Linux 6.8.0-124-generic, x86-64, glibc 2.39. GPU provenance is recorded for these experiments; CPU numerical and probe operations accompany GPU training. The earlier MAKE47 manuscript did state PyTorch 2.0, RTX 3070 and a CPU fallback; it did not identify the complete CUDA/cuDNN/driver environment.

The configuration file was created on 12 September 2026, with subsequent changes recorded on 12–13 September. These are internal specification records, not a public preregistration. Calibration informed some changes, including the dSprites likelihood. The pruning additions and full ladder were run during 14–17 September. The paired random-feature comparison and first accounting audit were retrospective revisions dated 18 September. The search replay, factorial threshold analysis, normalized ARD readout and bounded model validation were added on 19 September. MNIST and other previously examined datasets are not described as wholly unseen scientific conditions. “Evaluation condition” in the predictive analysis means withheld from fitting that predictor.

The logged code token 8c2c6c28224b3450 hashes only the common module and configuration, not the entire experimental implementation; the earlier trajectory records use 99cb5eb7404ff6e7. A new SHA-256 file manifest identifies the inspected snapshot. Appendix F maps numerical results to their source files and describes remaining provenance limitations.

5.4. Bounded New Model Validation

The archived MAKE49 models are not available as compatible checkpoints. To evaluate an actual ELBO, we train a separate MNIST grid at $\beta = 1$, seed 42, on the same 18,000/2000/3000 split and the same architecture, optimizer, 300-epoch limit and 30-epoch patience. This is a one-seed validation experiment on CPU, not a replacement for the five-seed CUDA comparison. Python 3.11.11, PyTorch 2.7.1, NumPy 2.3.2 and four CPU threads are used on macOS 26.6.2 (arm64). Its timings are not compared with GPU records.

Each width retains the checkpoint with the best posterior-mean proxy. We then evaluate

$$\hat{\mathcal{L}}_K = \frac{1}{nK} \sum_{i=1}^n \sum_{k=1}^K \log p_{\theta}(x_i | z_i^{(k)}) - \frac{1}{n} \sum_{i=1}^n \text{KL}\{q_{\phi}(z | x_i) \| p(z)\}, \quad z_i^{(k)} \sim q_{\phi}(z | x_i). \quad (5)$$

Here $K = 32$, the evaluation seed is 20260919, and $p_{\theta}(x | z) = \mathcal{N}(g_{\theta}(z), \frac{1}{2}I)$. The Gaussian normalizing constant is included. The same 64-dimensional normal draws are truncated for smaller widths. Monte Carlo SE is the sample SD of the 32 draw-wise validation means divided by $\sqrt{32}$; it measures evaluation noise, not training-seed uncertainty. Capacity is chosen separately by validation MSE, proxy and estimated ELBO, and choices are written before test evaluation. Thus the ELBO comparison selects capacity from *proxy-selected checkpoints*; it does not reselect every epoch using ELBO.

A separate 300-epoch, seed-42 MNIST run instruments the local hard-concrete (HC) pruning implementation. Its target comes from the new ordinary width-64 anchor. We retain native-model reconstruction, constraint, gate and multiplier trajectories and compare native and dimension-transferred outputs. This checks the behaviour of that local implementation in one condition; it is not an ARM replication or a validated cross-dataset pruning baseline. All new checkpoints are saved.

6. Results

6.1. Search Decisions and Canonical Costs

The historical MNIST expense contrast is structurally determined when the smallest qualifying width lies inside the initial window, $m_0 < m_a$, and no budget interruption occurs:

$$C_{\text{window}} - C_{\text{ascending}} = C_{\text{ref}} + \sum_{m^* < m \leq m_0} C_m \geq 0. \tag{6}$$

Both procedures already evaluate the anchor; its cost cancels. At the primary setting, $\hat{d} = 11.68$ gives $m_0 = 24$. The window requests widths 2, 4, 6, 8, 10, 12, 16, 20 and 24; ascending stops at 6. The historical ledger difference is $715.4 - 182.0 = 533.4 = 205.5 + (400.7 - 72.8)$ s (Appendix B.1). This is an accounting identity for those paths, not an empirical identification of the value of ID information.

Tables 3 and 4 evaluate policies with different stopping opportunities using uniform snapshot prices. ID-local requests nine candidates on MNIST, versus ten for the old window and four for ascending. All return the same width six. Canonical cold costs are 636, 672 and 177 s, respectively; these differ from historical ledger values because the prices are now uniform. The midpoint control requests six candidates and costs 279 s. Thus shortening the path alone does not establish an ID-specific gain.

Table 3. Retrospective search replay at $\beta = 1$, five seeds. Q : target attainment; exact: agreement with the full-grid smallest qualifying width; Δm : mean excess width. Calls include the anchor. Cost uses one canonical saved time per key plus the full reference bank for ID policies, in seconds; it is not new wall-clock timing. Gaussian likelihood except Bernoulli dSprites.

Policy	Width	Q	Exact	Δm	Calls	Cold cost
<i>MNIST</i>						
Full grid Q	6.0 ± 0.0	5/5	5/5	0.0	12.0	572
Ascending Q	6.0 ± 0.0	5/5	5/5	0.0	4.0	177
Legacy window	6.0 ± 0.0	5/5	5/5	0.0	10.0	672
ID local	6.0 ± 0.0	5/5	5/5	0.0	9.0	636
Midpoint local	6.0 ± 0.0	5/5	5/5	0.0	6.0	279
ID certified	6.0 ± 0.0	5/5	5/5	0.0	10.0	672
<i>Fashion-MNIST</i>						
Full grid Q	4.0 ± 0.0	5/5	5/5	0.0	12.0	645
Ascending Q	4.0 ± 0.0	5/5	5/5	0.0	3.0	209
Legacy window	6.4 ± 3.3	5/5	3/5	2.4	4.0	457
ID local	4.0 ± 0.0	5/5	5/5	0.0	3.0	416
Midpoint local	4.0 ± 0.0	5/5	5/5	0.0	7.0	407
ID certified	4.0 ± 0.0	5/5	5/5	0.0	3.0	416

Table 4. Retrospective search replay at $\beta = 1$, five seeds. Q: target attainment; exact: agreement with the full-grid smallest qualifying width; Δm : mean excess width. Calls include the anchor. Cost uses one canonical saved time per key plus the full reference bank for ID policies, in seconds; it is not new wall-clock timing. Gaussian likelihood except Bernoulli dSprites.

Policy	Width	Q	Exact	Δm	Calls	Cold cost
<i>dSprites</i>						
Full grid Q	20.0 ± 4.0	5/5	5/5	0.0	12.0	2563
Ascending Q	20.0 ± 4.0	5/5	5/5	0.0	9.0	1797
Legacy window	64.0 ± 0.0	5/5	0/5	44.0	4.0	999
ID local	20.0 ± 4.0	5/5	5/5	0.0	9.0	2092
Midpoint local	20.0 ± 4.0	5/5	5/5	0.0	4.0	941
ID certified	20.0 ± 4.0	5/5	5/5	0.0	9.0	2092
<i>CIFAR-10</i>						
Full grid Q	24.0 ± 0.0	5/5	5/5	0.0	10.0	428
Ascending Q	24.0 ± 0.0	5/5	5/5	0.0	4.0	157
Legacy window	131.2 ± 113.9	5/5	0/5	107.2	4.0	499
ID local	24.0 ± 0.0	5/5	5/5	0.0	4.0	498
Midpoint local	24.0 ± 0.0	5/5	5/5	0.0	5.0	213
ID certified	24.0 ± 0.0	5/5	5/5	0.0	4.0	498

Local search can reduce requests: on dSprites, midpoint-local takes four rather than nine ascending requests, with canonical costs 941 versus 1797 s and the same qualifying width in every primary seed. ID is rejected there, so ID-local follows ascending and additionally charges the bank. Across all 100 dataset–weight–seed conditions, ID-local and its certified variant match the full-grid minimum in all 100; midpoint-local matches in 98, with a maximum excess width of eight. All these new-policy outputs meet the validation target and none exceeds the original soft cost limit in the replay. These are conditional observations on a previously trained grid, not guarantees for new data.

6.2. Coefficient, Reliability and Reuse Sensitivity

At the primary MNIST setting, changing c from one to four leaves the selected width at six but changes ID-local requests from six to eleven and cold cost from 487 to 743 s. With the original reliability thresholds, the other three datasets use identical ascending fallback at every coefficient. Their invariance is therefore not evidence that the ID coefficient is robust.

At $c = 2$, relaxing $A_{\max}$ to 0.40 admits Fashion-MNIST and raises its primary request count from three to ten without improving the width. At 0.85 it also admits CIFAR-10; admitting dSprites additionally requires the permissive $R_{\max} = 0.50$, $A_{\max} = 2.00$ stress setting. Threshold choice thus changes whether ID actually participates. Tables A4 and A5 give full primary coefficient results and representative reliability contrasts; the supplement contains every factorial row.

Across all 10,000 sensitivity rows, quality is always met, but 155 outputs miss the full-grid minimum, with maximum excess width 56. All misses occur away from $\beta = 1$. They show why local stopping cannot be certified from apparent monotonicity at one setting. These rows reuse records and reference banks; 10,000 is a count of replay configurations, not independent experimental replicates. Table 5 separates cold use, warm use and one-time bank construction shared over 25 tasks.

Table 5. ID-local costs across five weights and five candidate seeds (25 tasks per dataset). Cold charges the bank each time; warm assumes an existing bank; batch allocates its training cost once across 25 tasks. All exclude unrecorded ID/evaluation overhead. Nonmono counts qualifying masks that change from pass to fail as width increases.

Dataset	Bank (s)	Cold (s)	Warm (s)	Batch (s)	Exact	Nonmono
MNIST	207.5	705.2	497.7	506.0	25/25	1/25
Fashion-MNIST	207.1	352.9	145.8	154.1	25/25	2/25
dSprites	295.8	2127.2	1831.4	1843.3	25/25	13/25
CIFAR-10	340.5	507.2	166.7	180.3	25/25	4/25

6.3. Recorded Selector Quality

Tables 6 and 7 retain ordinary-selector outcomes at $\beta = 1$, with sample SD over five training seeds. They report model widths, validation and test error and target attainment. Their timing ledgers are separated from the canonical replay prices. The inherited ARD and GECCO rows have been removed from this main comparison; their raw results and subsequent checks are implementation audits in Appendix C.2.

Table 6. Dimension selection at $\beta = 1$ (Gaussian likelihoods). Values are mean $\pm$ sample SD over five candidate-training seeds. m_{final} is the width of the evaluated ordinary VAE. MSE columns are multiplied by 10^3 ; Q counts validation target attainment. Historical timing ledgers are separated in Appendix B.1; replay costs use a single canonical snapshot.

Method	m_{final}	Val. MSE	Test MSE	Q
<i>MNIST</i>				
Grid (MSE)	12.8 ± 6.4	20.98 ± 0.24	20.50 ± 0.26	5/5
FONDUE	7.0 ± 0.0	21.93 ± 0.77	21.54 ± 0.78	5/5
FONDUE + Q	8.0 ± 0.0	21.72 ± 0.50	21.33 ± 0.55	5/5
Ascending + Q	6.0 ± 0.0	22.44 ± 0.15	22.05 ± 0.17	5/5
Legacy ID window	6.0 ± 0.0	22.44 ± 0.15	22.05 ± 0.17	5/5
<i>Fashion-MNIST</i>				
Grid (MSE)	22.8 ± 15.6	18.33 ± 0.20	18.08 ± 0.19	5/5
FONDUE	6.8 ± 0.4	19.29 ± 0.16	19.01 ± 0.16	5/5
FONDUE + Q	7.6 ± 0.9	19.00 ± 0.33	18.75 ± 0.34	5/5
Ascending + Q	4.0 ± 0.0	19.38 ± 0.24	19.06 ± 0.25	5/5
Legacy ID window	6.4 ± 3.3	19.03 ± 0.61	18.73 ± 0.57	5/5

Table 7. Dimension selection at $\beta = 1$ (Bernoulli for dSprites; Gaussian for CIFAR-10). Values are mean $\pm$ sample SD over five candidate-training seeds. m_{final} is the width of the evaluated ordinary VAE. MSE columns are multiplied by 10^3 ; Q counts validation target attainment. Historical timing ledgers are separated in Appendix B.1; replay costs use a single canonical snapshot.

Method	m_{final}	Val. MSE	Test MSE	Q
<i>dSprites</i>				
Grid (MSE)	57.6 ± 14.3	1.16 ± 0.01	1.15 ± 0.01	5/5
FONDUE	4.2 ± 0.4	5.50 ± 1.31	5.42 ± 1.28	0/5
FONDUE + Q	20.0 ± 4.0	1.26 ± 0.05	1.24 ± 0.05	5/5
Ascending + Q	20.0 ± 4.0	1.26 ± 0.05	1.24 ± 0.05	5/5
Legacy ID window	64.0 ± 0.0	1.17 ± 0.02	1.16 ± 0.03	5/5
<i>CIFAR-10</i>				
Grid (MSE)	140.8 ± 53.5	13.83 ± 0.04	14.07 ± 0.04	5/5
FONDUE	14.4 ± 0.5	17.79 ± 0.30	18.10 ± 0.30	0/5
FONDUE + Q	24.0 ± 0.0	15.12 ± 0.14	15.36 ± 0.13	5/5
Ascending + Q	24.0 ± 0.0	15.12 ± 0.14	15.36 ± 0.13	5/5
Legacy ID window	131.2 ± 113.9	14.01 ± 0.13	14.25 ± 0.14	5/5

Ordinary FONDUE returns approximately four dimensions on dSprites but has validation MSE 0.00550, versus 0.00126 for ascending search. Normal termination does not imply quality attainment. FONDUE+Q includes one budget-exceeded dSprites trial that

still returned a model; that model remains in the summary. Independent outcome counts are in Appendix A.1. The legacy ID rows for Fashion-MNIST, dSprites and CIFAR-10 use coarse-MSE fallback; the high CIFAR-10 width variability belongs to that fallback. Only CIFAR-10 is a natural colour-image benchmark here.

6.4. Representation and Sample Dependence of ID

Table 8 gives the complete reference-bank summary, including Fashion-MNIST. Only MNIST passes both checks. Because the same bank is reused, five identical pass/fail decisions do not constitute five independent replications of ID estimation.

Table 8. ID-reference AE widths and TwoNN estimates (mean $\pm$ SD over three reference seeds); stability R and estimator disagreement A . Limits are $R \leq 0.15$ and $A \leq 0.25$. The same reference bank is reused across candidate seeds and β .

Dataset	Width: estimate	R	A
MNIST	55: 10.93 ± 0.08 ; 111: 11.78 ± 0.12 ; 221: 12.32 ± 0.09	0.118	0.150
Fashion-MNIST	49: 9.91 ± 0.06 ; 98: 10.45 ± 0.04 ; 197: 10.83 ± 0.05	0.088	0.361
dSprites	30: 3.96 ± 0.48 ; 61: 3.91 ± 1.12 ; 122: 2.10 ± 0.02	0.474	1.816
CIFAR-10	81: 13.60 ± 0.09 ; 162: 13.44 ± 0.71 ; 325: 12.34 ± 0.50	0.094	0.834

Figure 1 separately shows the sample-size and PCA analyses. At 1000 examples the two estimators agree within the specified 0.25 relative tolerance on all four raw-input datasets; at 10,000, only MNIST does. These are raw-input checks, distinct from the AE-latent checks used by the selector. On dSprites, input TwoNN is 7.70 in this diagnostic, PCA estimates range from 2.68 to 4.32, and individual AE estimates range from 2.08 to 4.56. The combined range is about 3.7-fold; the corresponding MNIST range is about 1.7-fold. These are representation-dependent estimates, not competing measurements of a known ground truth.

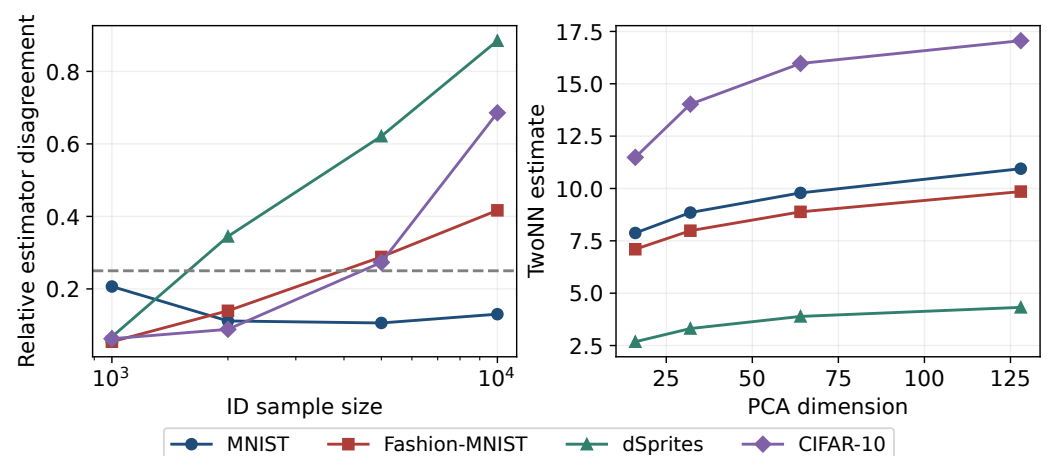


Figure 1. Left: $|d_{\text{TwoNN}} - d_{\text{MLE}}|/d_{\text{TwoNN}}$ on raw inputs as estimation sample size changes; dashed line is 0.25. Right: TwoNN after PCA preprocessing. PCA rank, sample size and representation are distinct interventions; AE results are in Table 8.

6.5. Dependence on Regularisation and Quality Context

Define the reported range for method a and dataset d by

$$S_{a,d} = \max_{\beta \in \mathcal{B}} \bar{m}_{a,d}(\beta) - \min_{\beta \in \mathcal{B}} \bar{m}_{a,d}(\beta), \quad \bar{m}_{a,d}(\beta) = \frac{1}{5} \sum_{s=1}^5 m_{a,d,s}(\beta), \quad \mathcal{B} = \{0.25, 0.5, 1, 2, 4\}. \quad (7)$$

This is the range of seed means of raw returned dimensions, not the mean of within-seed ranges. For the FONDUE-VAR row all five outputs are defined at each setting, including any raw zero. It is a descriptive range on the tested ladder, not a new parameter-free robustness index.

Table 9. Range of seed-mean returned dimensions across $\beta \in \{0.25, 0.5, 1, 2, 4\}$. Pruning adaptations are restricted to the implementation audit. Fixed 32 is a design constant; fallback spans are not ID robustness. Gaussian likelihood except Bernoulli dSprites.

Method	MNIST	Fashion	dSprites	CIFAR-10
Grid (MSE)	14.8	20.8	40.0	91.2
Coarse search	33.6	37.2	0.0	214.4
Fixed 32	0.0	0.0	0.0	0.0
FONDUE	3.8	0.8	0.8	8.8
FONDUE-VAR (mixed kept)	50.6	63.0	106.2	68.8
Ascending + Q	6.0	3.6	15.2	52.8
FONDUE + Q	3.6	1.6	28.0	44.8
Legacy ID window	6.0	37.2	0.0	214.4

Ranges differ by method and dataset, with clear ordering reversals. For example, ascending versus FONDUE+Q ranges are 6.0 versus 3.6 on MNIST, but 15.2 versus 28.0 on dSprites. The legacy ID window’s CIFAR-10 range of 214.4 is inherited from fallback. A fixed dimension has zero range by design; dSprites ID fallback has zero range because coarse search always returns 64. Neither demonstrates useful selection robustness. We do not average ranges across differently scaled grids or assert a transferable ranking.

The ARD call did not vary its selection-stage weight and GECCO varied an anchor-derived constraint rather than β directly. Both belong to the implementation audit, not Figure 2’s ordinary-VAE regularisation comparison.

An AU-threshold intervention holds the checkpoint fixed and recomputes Equation (4). A quality-margin intervention instead changes which cached models satisfy the target. Neither is the same as changing training β . We report these readout/quality analyses separately (Section 6.7 and Appendix C.3). No sensitivity claim is made for untested FONDUE stopping thresholds or ARD cumulative relevance fractions.

6.6. Learning Curves, Test Evaluation and Selection Criteria

Figure 3 uses actual early-stopped cache records selected by ascending-Q at the primary setting, including MNIST width six and CIFAR-10. Each run is shown only over its recorded epochs and the proxy-selected checkpoint is marked. These cache records contain validation curves but no epoch-wise train curve; no missing train measurements are imputed. The separate new MNIST run supplies matched train/validation curves for width six (Appendix E).

The saved trajectory experiment supplies train/validation curves for MNIST widths 12, 32 and 64 and dSprites widths 6, 16 and 32, with five seeds each (Figure 4). Training MSE is evaluated in deterministic mode on a fixed 2000-example training subset sampled with seed 20260919. Both train and validation improve, but validation error remains higher. None of these 30 runs passes the stringent final 50-epoch plateau check; completing 300 epochs is not proof of convergence. These curves cover representative settings, not all four datasets.

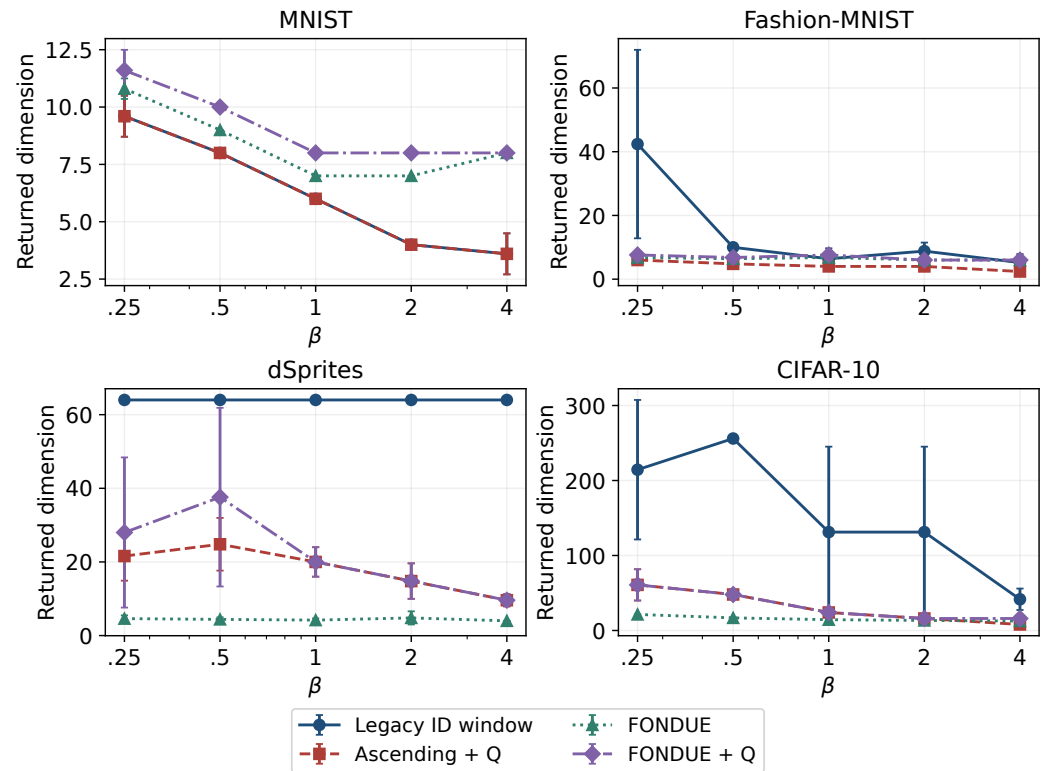


Figure 2. Raw returned dimension across the five contexts, mean $\pm$ SD over five candidate seeds. Gaussian likelihood except Bernoulli dSprites. These ordinary VAE selectors change training β . The legacy ID window falls back on Fashion-MNIST, dSprites and CIFAR-10 throughout the ladder.

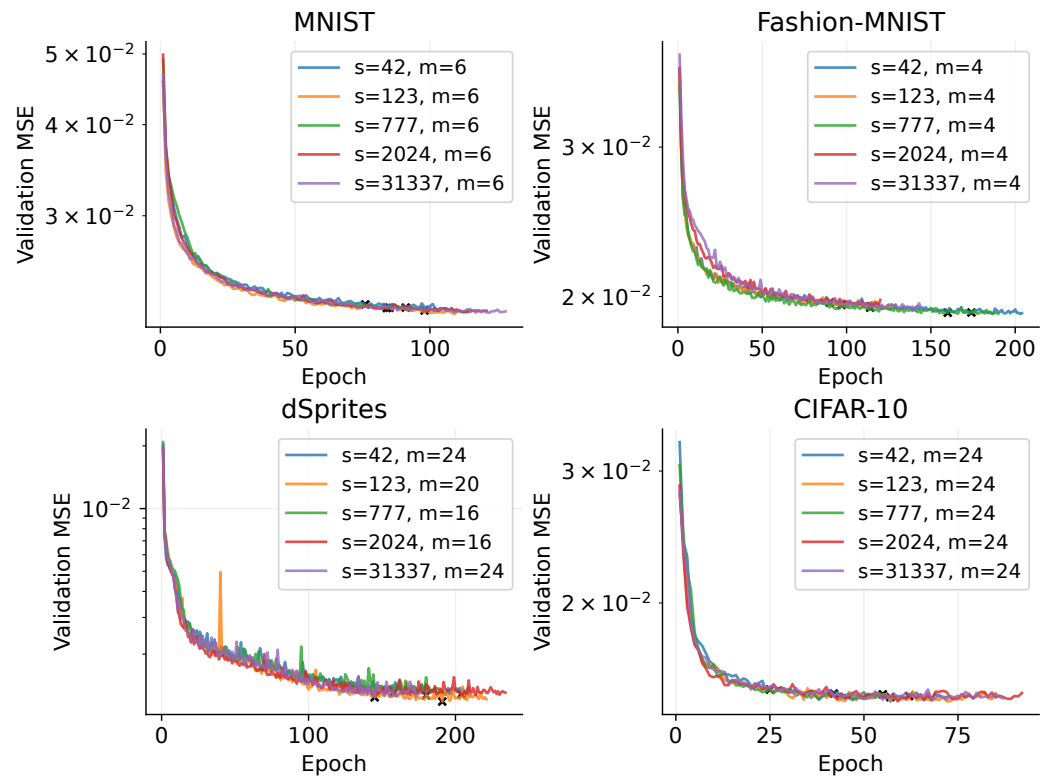


Figure 3. Actual selected-candidate validation MSE at $\beta = 1$, five seeds, Gaussian likelihood except Bernoulli dSprites. Widths are chosen by ascending-Q; black crosses mark proxy-selected epochs. Curves end at the recorded stopping time. These are not the fixed-300-epoch diagnostic runs in Figure 4.

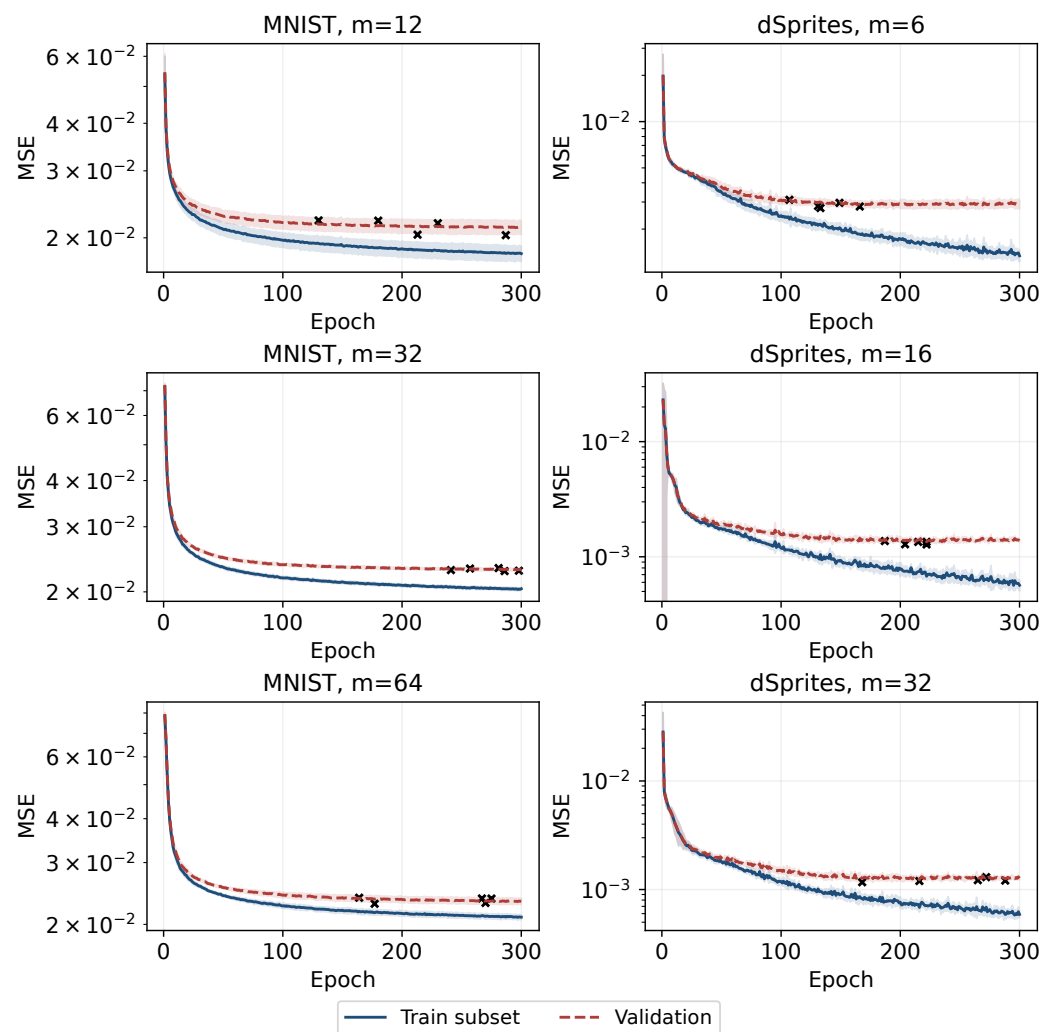


Figure 4. Train and validation MSE, means with ± 1 SD bands over five seeds. MNIST uses Gaussian and dSprites Bernoulli likelihood; $\beta = 1$. Black crosses mark each run's checkpoint selected by the validation ELBO proxy. These diagnostic runs have no early stopping.

Table 10. Selection-criterion comparison at $\beta = 1$, five seeds; test MSE $\times 10^3$ and probe test accuracy (%). Probes use class labels (shape for dSprites); only Grid (probe) uses them for capacity selection. Time is the historical ledger including anchor completion, not the canonical replay price. Recorded probe time is charged to that selector. Other displayed probe accuracies are auxiliary evaluations with uncharged costs. Likelihoods follow Table 2.

Dataset	Criterion	m	Test MSE	Accuracy	Time (s)
MNIST	Grid (MSE)	12.8 ± 6.4	20.50 ± 0.26	89.4 ± 0.9	628 ± 35
MNIST	Grid (ELBO proxy)	10.0 ± 2.0	20.54 ± 0.25	89.3 ± 0.7	628 ± 35
MNIST	Grid (probe)	51.2 ± 13.4	23.01 ± 0.55	92.7 ± 0.5	696 ± 36
Fashion-MNIST	Grid (MSE)	22.8 ± 15.6	18.08 ± 0.19	75.5 ± 2.6	645 ± 42
Fashion-MNIST	Grid (ELBO proxy)	15.6 ± 8.0	18.12 ± 0.27	74.4 ± 2.6	645 ± 42
Fashion-MNIST	Grid (probe)	57.6 ± 8.8	18.99 ± 0.46	78.4 ± 0.6	750 ± 40
dSprites	Grid (MSE)	57.6 ± 14.3	1.15 ± 0.01	47.7 ± 2.3	2565 ± 82
dSprites	Grid (ELBO proxy)	57.6 ± 14.3	1.15 ± 0.01	47.7 ± 2.3	2565 ± 82
dSprites	Grid (probe)	64.0 ± 0.0	1.16 ± 0.03	49.8 ± 3.0	2589 ± 84
CIFAR-10	Grid (MSE)	140.8 ± 53.5	14.07 ± 0.04	44.3 ± 2.6	428 ± 16
CIFAR-10	Grid (ELBO proxy)	192.0 ± 45.3	14.11 ± 0.08	46.1 ± 1.1	428 ± 16
CIFAR-10	Grid (probe)	256.0 ± 0.0	14.30 ± 0.12	47.3 ± 1.0	574 ± 15

Table 10 provides the previously omitted grid choices based on MSE, the ELBO proxy and a linear probe. Different objectives can select different widths and reconstruction errors. Test probe accuracy is an auxiliary supervised outcome; it does not measure unsupervised reconstruction quality. This comparison retains the same splits and cached candidates, with labelled validation selection explicitly stated.

The separate new MNIST comparison in Table 11 evaluates an actual expected log likelihood with the fixed Monte Carlo protocol in Section 5.4. It is not reconstructed from cached MSE and KL. Validation MSE selects width 12, the proxy selects 12, and Monte Carlo ELBO selects 12. Agreement in this run does not make the proxy an ELBO estimator. The complete grid scores, MC errors and selected epochs appear in Appendix E. One new training seed does not establish a dataset-level advantage of any criterion.

Table 11. New MNIST seed-42 capacity selection at $\beta = 1$, Gaussian variance $1/2$. All criteria choose among the same twelve proxy-selected checkpoints. MSE is multiplied by 10^3 ; ELBO is mean nats/example and its $\pm$ value is Monte Carlo SE ($K = 32$), not a training-seed SD. Test is evaluated after validation choices are frozen.

Criterion	Width	Val. MSE	Test MSE	MC ELBO $\pm$ MC SE
MSE	12	20.98	20.43	-478.067 ± 0.012
ELBO proxy	12	20.98	20.43	-478.067 ± 0.012
MC ELBO	12	20.98	20.43	-478.067 ± 0.012

6.7. Activity Readout: Trajectories, Thresholds and Prediction

Figure 5 and Table 12 compare the two counts on identical checkpoints. At MNIST width 64, epoch-five variable types give 9.8 ± 3.8 and AU gives 49.4 ± 8.1 ; at epoch 300 both give 6.4 ± 0.5 . This example distinguishes mean agreement from between-seed variability. The table also reports within-run changes from 50 to 300 epochs. Their magnitude depends on the condition, so no general stability superiority is inferred from the final agreement.

Table 12. Variable-type count V (active plus mixed) and AU count A , five seeds, $\beta = 1$, Gaussian MNIST and Bernoulli dSprites. Mean $\pm$ SD quantify between-seed variation. The last column gives mean absolute within-run changes from epoch 50 to 300 for V/A , not a temporal variance.

Dataset	m	V_5	A_5	V_{300}	A_{300}	Change
MNIST	12	7.2 ± 0.8	8.0 ± 1.6	6.4 ± 0.5	6.4 ± 0.5	0.0/0.0
MNIST	32	8.2 ± 1.5	20.8 ± 4.2	6.0 ± 0.0	6.0 ± 0.0	0.0/0.0
MNIST	64	9.8 ± 3.8	49.4 ± 8.1	6.4 ± 0.5	6.4 ± 0.5	0.4/2.6
dSprites	6	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	0.0/0.0
dSprites	16	16.0 ± 0.0	16.0 ± 0.0	14.2 ± 0.4	13.4 ± 0.5	1.8/1.8
dSprites	32	32.0 ± 0.0	31.8 ± 0.4	14.2 ± 1.1	14.4 ± 1.1	12.4/4.2

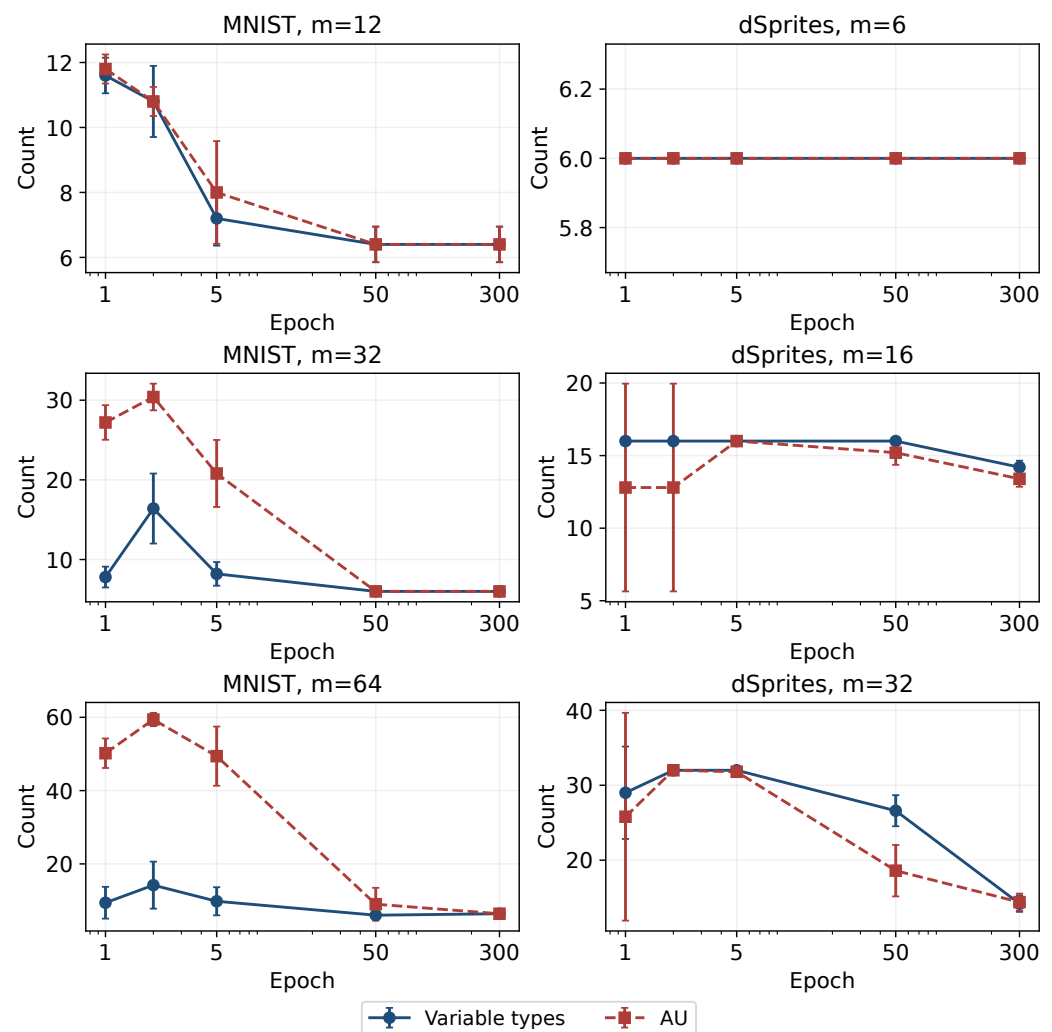


Figure 5. Variable-type count (active plus mixed) and AU from the same checkpoints at epochs 1, 2, 5, 50 and 300; means and seed SDs, with lines connecting the observed checkpoints only. $\beta = 1$, Gaussian MNIST and Bernoulli dSprites. Constant full-capacity counts are not evidence of diagnostic stability.

The threshold-only comparison in Figure 6 uses fixed $\beta = 1$ checkpoints. On CIFAR-10, $\delta = 0.001$ counts every coordinate at every tested grid width, whereas $\delta = 0.1$ counts far fewer at large widths. The inference “no unused coordinates” therefore depends on this threshold. Normalising the threshold changes the diagnostic definition; it does not establish a dimension estimate.

The predictive task asks whether the next smaller grid candidate meets the same seed’s quality target. Each row is one validation-selected checkpoint at a nonminimal grid

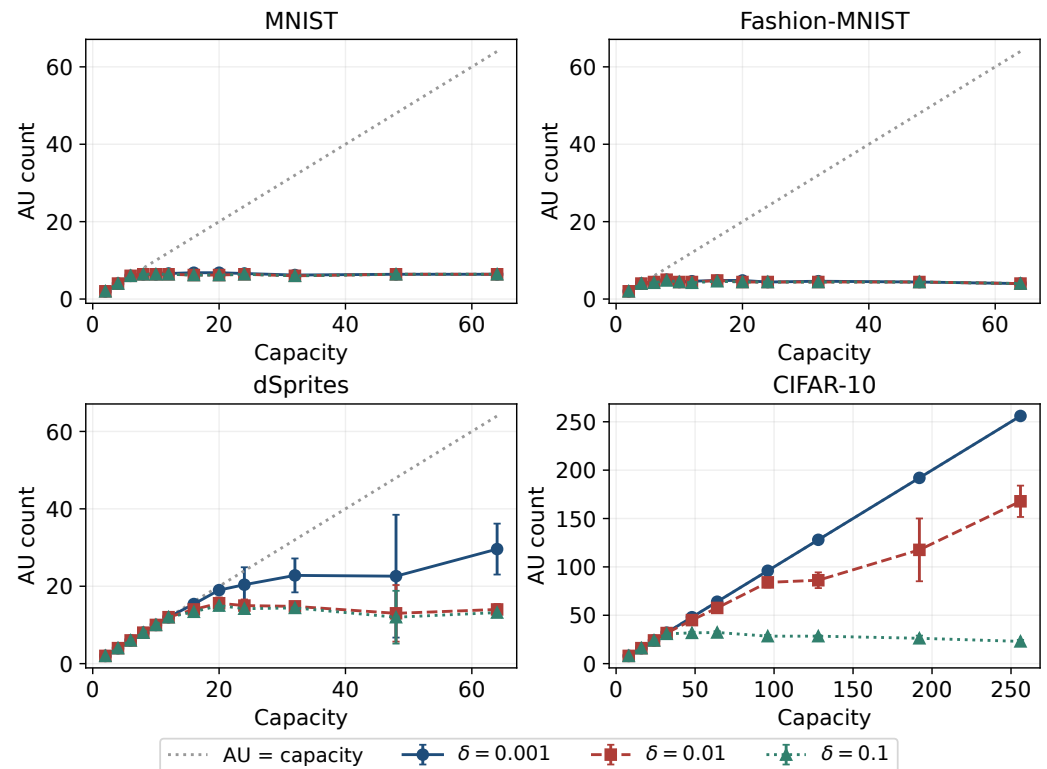


Figure 6. AU under three thresholds on the same $\beta = 1$ candidates; mean $\pm$ seed SD. Gaussian likelihood except Bernoulli dSprites. The diagonal is capacity. This is a readout-threshold intervention, unlike the training/context intervention in Figure 2.

width. Baseline features are validation MSE, KL, the ELBO proxy, width, width/input-ID ratio, and the slope of validation MSE over the last 20 recorded epochs. The three added AU features are AU, AU/width and AU/input-ID. Standardisation is fitted only on development rows. An L_2 logistic predictor is fitted on MNIST (55 rows), with $C = 10$ chosen on a three-seed/two-seed development split. Evaluation uses dSprites (55 rows), Fashion-MNIST (55) and CIFAR-10 (45), each with five seeds. The configured MNIST-CNN predictive condition was not executed.

The development split uses seeds 42, 123 and 777 for the inner fit and 2024 and 31337 for selecting C , then refits on all five. Predictors use up to 5000 iterations. Each evaluation dataset-seed block is scored over its complete set of nonminimal grid widths; checkpoint rows are not resampled as independent observations.

The original internally specified analysis reported an AU-minus-baseline AU-ROC improvement of $+0.108$, with a 2000-replicate seed-record bootstrap 95% interval $[+0.059, +0.157]$. A subsequently added control used three independent standard-normal features and one noise realisation (generator seed 7); its separately reported improvement was $+0.098$ with interval $[-0.025, +0.227]$. Overlap of these intervals does not test their difference or establish equivalence. These archived values are retained as the original analysis, not as a validated positive finding: its AUROC calculation used probabilities that can saturate numerically under the shift between datasets.

Table 13. Retrospective paired AU comparison at $\beta = 1$, using logistic decision scores to avoid saturated-probability ties. Values are means of five seed-specific AUROCs, each evaluated over all nonminimal capacities in that seed. Random features are three standard-normal columns with generator seed 7. All methods use the same rows and labels.

Evaluation condition	Rows	Base	Base + AU	Base + random	AU – random
dSprites	55	0.950	0.858	0.898	-0.041
Fashion-MNIST	55	0.980	1.000	0.880	+0.120
CIFAR-10	45	1.000	1.000	1.000	+0.000

Baseline AUROC is already 0.950, 0.980 and 1.000 on dSprites, Fashion-MNIST and CIFAR-10, respectively. In particular, CIFAR-10 has no upward AUROC headroom. The result concerns additional value on this specific, near-ceiling prediction task, not the information content of AU in general.

For this revision we reconstruct both predictors at explicitly fixed $\beta = 1$, restrict rows to the declared grid, and compute AUROC from logistic decision scores. The original cache-reading routine did not filter β or restrict the grid, so rerunning it after cache expansion would also be ambiguous. Decision scores preserve ranks that are lost when sigmoid probabilities round to exactly one. Saturation counts appear in Appendix C.3. This is a corrected retrospective analysis, not an exact replay of archived probabilities. The AU-minus-baseline difference is now -0.024 , with interval $[-0.102, +0.027]$. We also compare AU and random predictors on the same dataset-seed blocks. The paired mean difference is $+0.026$, with a retrospective hierarchical-bootstrap 95% interval $[-0.040, +0.133]$ (2000 replicates). Each replicate resamples the three evaluation datasets and then the five seed blocks within each selected dataset; all capacities and both predictors remain paired within a block. Models are held fixed, so this interval does not cover development-model fitting uncertainty. Only three evaluation datasets are available, limiting precision of across-condition inference. A secondary analysis averages the comparator over 100 noise realisations (seeds 0–99), giving a paired difference -0.007 and interval $[-0.064, +0.039]$ under the same resampling. Neither analysis establishes an AU-specific benefit in these evaluation conditions. We do not claim that AU has no information, that the predictors are equivalent, or that lack of significance proves insufficient power.

6.8. Implementation Audit and Degenerate Outputs

Two scale inconsistencies were identified in local implementations. First, the earlier VAE loss averaged KL over latent dimensions while summing reconstruction over input dimensions. Algebraically this gives $\beta_{\text{effective}} = \beta_{\text{nominal}}/m$. The image results in this paper come from the corrected sum-over-latents convention. The algebra does not imply that complete optimisation trajectories or AU counts are reproduced simply by relabelling β ; the earlier “within 1.7 units” claim is not used here.

Second, an intermediate GECO adaptation combined per-pixel-mean constraint values with other terms on incompatible scales. The corrected implementation uses summed squared error and multiplies the per-pixel target by the input dimension. Either sum or mean conventions can be valid if the objective, target and multiplier scales are consistent. The observed behaviour of this hard-concrete adaptation is not attributed to the original ARM implementation.

FONDUE-VAR’s active-only setting returning zero is a separate phenomenon, not a third sum/mean defect. At MNIST’s one-epoch setting it returns zero on all five trials; no final VAE exists for these outputs. Table 14 shows both settings and their quality/validity counts.

Table 14. FONDUE-VAR at $\beta = 1$: both keep-mixed settings, valid-model and quality counts, test MSE $\times 10^3$, and recorded training time. Raw zero outputs remain in the dimension summary; nonexistent models have no MSE. Five seeds; likelihoods follow Table 2.

Dataset	Mixed kept	Raw m	Valid	Q	Test MSE	Time (s)
MNIST	yes	9.8 ± 8.5	5/5	5/5	22.53 ± 0.89	113 ± 22
MNIST	no	0.0 ± 0.0	0/5	0/5	—	63 ± 10
Fashion-MNIST	yes	14.4 ± 3.8	5/5	5/5	18.96 ± 0.39	91 ± 11
Fashion-MNIST	no	2.0 ± 0.0	5/5	0/5	25.27 ± 0.06	120 ± 18
dSprites	yes	26.0 ± 49.5	4/5	1/5	9.53 ± 9.62	412 ± 118
dSprites	no	1.2 ± 1.6	2/5	0/5	7.71 ± 0.02	278 ± 49
CIFAR-10	yes	34.2 ± 1.3	5/5	5/5	14.40 ± 0.15	103 ± 13
CIFAR-10	no	3.2 ± 0.8	5/5	0/5	31.65 ± 2.68	96 ± 13

7. Discussion

Search design changes what a cost comparison can identify. Exhausting a lower ID window includes work already sufficient for ascending search; its higher expense is largely determined by construction. The new replay instead exposes the trade-off between fewer candidate requests and certifying the smallest qualifying width. A no-ID local control can also save requests. The results therefore do not attribute every reduction from full-grid search to ID information.

ID-local matches the grid minimum at the inherited default throughout the recorded ladder, but relies on fallback in three datasets. The factorial analysis shows both decision changes at reliability thresholds and nonminimal outputs at other settings. Shared reference banks, one split, retrospective policy design and a small dataset collection limit generalization. Acquisition and reuse costs answer different deployment questions and cannot be collapsed into one universal ranking.

Regularisation ranges and AU-threshold responses are different measurements. Some large dimension ranges reflect fallback, grid scale or a changing quality target. A constant output can be fixed by design or by an unvaried implementation argument. Thus the evidence supports method- and dataset-dependent responses, not a universal robustness ranking. The archived positive AU-versus-baseline result is not robust to preserving logistic-score rankings instead of rounded probabilities. The corrected predictive comparison does not establish an AU-specific advantage over the matched random-feature controls.

The pruning evidence remains deliberately outside the main comparison. Normalizing the stored ARD perturbations corrects one identifiable readout error, but does not validate the prior-update procedure or recover an infinitesimal Jacobian. The native HC trajectory supplies one new implementation diagnostic, not broad source-method validation. The new MC ELBO comparison likewise covers one MNIST seed and proxy-selected checkpoints; it does not complete a four-dataset, five-seed ELBO study. Exact ARM/source-ARD reproduction, an independent reference-bank replication, missing natural-image training curves and complete end-to-end timing remain open. The contribution is thus a reproducible comparison of specified search policies and diagnostic assumptions, not a definitive ranking of established pruning methods.

The synthetic experiments in Appendix D illustrate scale dependence and the effect of zeroing low-variance coordinates. Their counts are observations under a particular loss and coordinate system. They do not show that topology predicts the number of active axes, and their preprocessing prevents treating them as clean estimates of held-out generalisation.

8. Conclusions

We separated latent-width search decisions from diagnostic readouts and implementation history. The historical fixed-window expense contrast does not by itself test the value of ID information. Retrospective local-search and matched-control replays quantify candi-

date savings, reference acquisition costs and gaps from the full-grid minimum. Reliability thresholds alter which datasets use ID, and some local policies miss qualifying smaller widths. The paired AU analysis does not establish an AU-specific benefit on its near-ceiling prediction task. A bounded new ELBO experiment and corrected relevance audit improve measurement validity while leaving source-method validation incomplete. Claims about efficient capacity selection therefore depend on explicit search rules, quality targets, cost units and implementation evidence.

Author Contributions: Conceptualization, C.O., M.D. and K.J.; methodology, C.O. and K.J.; software, C.O. and M.D.; validation, C.O., M.D. and K.J.; formal analysis, C.O. and K.J.; investigation, C.O., M.D. and K.J.; writing—original draft preparation, C.O. and K.J.; writing—review and editing, C.O., M.D. and K.J.; supervision and project administration, K.J.

Funding: This research was funded by JSPS KAKENHI Grant Numbers JP24K15115 and JP26K14984, by the Tokyo City University Priority Research [Intelligent Technology Optimization Research Unit: ORBIT], by a Cooperative Research Project of the Research Institute of Electrical Communication (RIEC), Tohoku University, and by a Research Grant from the Telecommunications Advancement Foundation (TAF).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The image datasets are publicly available from their cited sources. The accompanying reviewer supplement `MAKE51_reproducibility.zip` contains the configuration, original result records, replay tables, analysis code, dependencies, split indices, new MNIST checkpoints and a SHA-256 manifest. Extract the archive and follow its `README.md`; Appendix F maps its contents. Historical MAKE49 checkpoints are unavailable. The archive is supplied with this revision; it is not described as a publicly deposited dataset and has no archive DOI.

Acknowledgments: Generative-AI assistance was used for experimental-code drafting and refactoring and English-language editing in the earlier manuscript preparation. OpenAI Codex assisted with the present record audit, reaggregation, additional model validation, figure preparation and manuscript revision. Responsibility for the scientific content rests with the authors.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Quality Targets and Outcome Accounting

564

Table A1. Quality anchors and targets, in units of 10^{-3} MSE, mean $\pm$ SD over five seeds. Each decision uses its own seed-specific target, not the displayed mean. The relative margin is 10%; the absolute floor is inactive. Gaussian likelihood except Bernoulli dSprites.

Dataset	β	Anchor MSE	Target $T_{D,s}$
MNIST	0.25	15.843 $\pm$ 0.562	17.428 $\pm$ 0.618
MNIST	0.5	19.202 $\pm$ 0.497	21.122 $\pm$ 0.546
MNIST	1	23.712 $\pm$ 0.532	26.084 $\pm$ 0.585
MNIST	2	30.061 $\pm$ 0.560	33.068 $\pm$ 0.617
MNIST	4	38.977 $\pm$ 0.673	42.875 $\pm$ 0.740
Fashion-MNIST	0.25	15.591 $\pm$ 0.678	17.150 $\pm$ 0.746
Fashion-MNIST	0.5	17.408 $\pm$ 0.513	19.148 $\pm$ 0.564
Fashion-MNIST	1	19.679 $\pm$ 0.162	21.647 $\pm$ 0.179
Fashion-MNIST	2	22.416 $\pm$ 0.527	24.658 $\pm$ 0.580
Fashion-MNIST	4	25.967 $\pm$ 0.587	28.564 $\pm$ 0.645
dSprites	0.25	1.015 $\pm$ 0.032	1.117 $\pm$ 0.036
dSprites	0.5	1.040 $\pm$ 0.032	1.144 $\pm$ 0.035
dSprites	1	1.170 $\pm$ 0.024	1.287 $\pm$ 0.026
dSprites	2	1.453 $\pm$ 0.038	1.598 $\pm$ 0.041
dSprites	4	2.144 $\pm$ 0.064	2.358 $\pm$ 0.070
CIFAR-10	0.25	9.407 $\pm$ 0.610	10.348 $\pm$ 0.671
CIFAR-10	0.5	11.151 $\pm$ 0.200	12.266 $\pm$ 0.220
CIFAR-10	1	14.064 $\pm$ 0.108	15.470 $\pm$ 0.119
CIFAR-10	2	17.806 $\pm$ 0.262	19.587 $\pm$ 0.288
CIFAR-10	4	22.399 $\pm$ 0.286	24.638 $\pm$ 0.314

Appendix A.1. Independent Outcome Counts

565

The following tables separate the original combined state from flags and final-model quality. A budget flag can coexist with ID fallback. The pruning wrapper did not enforce the ordinary selector budget during training, so its zero budget-flag count is not evidence of equal budget control. All valid returned models, including those from budget-exceeded trials, contribute to the primary means. Degenerate outputs contribute to raw-count and cost summaries but not to MSE.

566

567

568

569

570

571

Table A2. Outcome audit at $\beta = 1$; all counts have denominator five. S: recorded success; I: ID unreliable; B: budget exceeded; D: degenerate; N: no compact alternative. FB is the fallback flag; budget, validation quality Q , and compression of the final VAE are independently recomputed or read from flags.

Dataset	Method	State	FB	Budget	Q	Compact
MNIST	Grid (MSE)	S5	0	0	5	5
MNIST	FONDUE	S5	0	0	5	5
MNIST	FONDUE + Q	S5	0	0	5	5
MNIST	GECO adaptation	S5	0	0	5	5
MNIST	ARD adaptation	S5	0	0	5	5
MNIST	Ascending + Q	S5	0	0	5	5
MNIST	Legacy ID window	S5	0	0	5	5
MNIST	FONDUE-VAR (mixed kept)	S5	0	0	5	5
MNIST	FONDUE-VAR (active only)	D5	0	0	0	0
Fashion-MNIST	Grid (MSE)	S5	0	0	5	5
Fashion-MNIST	FONDUE	S5	0	0	5	5
Fashion-MNIST	FONDUE + Q	S5	0	0	5	5
Fashion-MNIST	GECO adaptation	S5	0	0	5	5
Fashion-MNIST	ARD adaptation	S5	0	0	4	5
Fashion-MNIST	Ascending + Q	S5	0	0	5	5
Fashion-MNIST	Legacy ID window	I5	5	0	5	5
Fashion-MNIST	FONDUE-VAR (mixed kept)	S5	0	0	5	5
Fashion-MNIST	FONDUE-VAR (active only)	S5	0	0	0	5

Table A3. Outcome audit at $\beta = 1$; all counts have denominator five. S: recorded success; I: ID unreliable; B: budget exceeded; D: degenerate; N: no compact alternative. FB is the fallback flag; budget, validation quality Q , and compression of the final VAE are independently recomputed or read from flags.

Dataset	Method	State	FB	Budget	Q	Compact
dSprites	Grid (MSE)	S5	0	0	5	1
dSprites	FONDUE	S5	0	0	0	5
dSprites	FONDUE + Q	B1, S4	0	1	5	5
dSprites	GECO adaptation	S5	0	0	5	0
dSprites	ARD adaptation	S5	0	0	0	5
dSprites	Ascending + Q	S5	0	0	5	5
dSprites	Legacy ID window	I5	5	0	5	0
dSprites	FONDUE-VAR (mixed kept)	D1, S4	0	0	1	3
dSprites	FONDUE-VAR (active only)	D3, S2	0	0	0	2
CIFAR-10	Grid (MSE)	S5	0	0	5	5
CIFAR-10	FONDUE	S5	0	0	0	5
CIFAR-10	FONDUE + Q	S5	0	0	5	5
CIFAR-10	GECO adaptation	S5	0	0	2	5
CIFAR-10	ARD adaptation	S5	0	0	0	5
CIFAR-10	Ascending + Q	S5	0	0	5	5
CIFAR-10	Legacy ID window	I5	5	0	5	3
CIFAR-10	FONDUE-VAR (mixed kept)	S5	0	0	5	5
CIFAR-10	FONDUE-VAR (active only)	S5	0	0	0	5

Appendix B. Search Sensitivity and Timing Lineage

These tables use the retrospective policy in Algorithm 1. They do not redefine the original experiments.

Table A4. Coefficient sensitivity for ID-local replay at $\beta = 1$, with $R \leq 0.15$ and $A \leq 0.25$. All five coefficients are reported; rejected datasets use the same ascending fallback. Five seeds; likelihoods as in Table 2.

Dataset	c	Width	Q	Exact	Calls	Cold cost (s)
MNIST	1	6.0 ± 0.0	5/5	5/5	6.0	487
MNIST	1.5	6.0 ± 0.0	5/5	5/5	8.0	584
MNIST	2	6.0 ± 0.0	5/5	5/5	9.0	636
MNIST	3	6.0 ± 0.0	5/5	5/5	11.0	743
MNIST	4	6.0 ± 0.0	5/5	5/5	11.0	743
Fashion-MNIST	1	4.0 ± 0.0	5/5	5/5	3.0	416
Fashion-MNIST	1.5	4.0 ± 0.0	5/5	5/5	3.0	416
Fashion-MNIST	2	4.0 ± 0.0	5/5	5/5	3.0	416
Fashion-MNIST	3	4.0 ± 0.0	5/5	5/5	3.0	416
Fashion-MNIST	4	4.0 ± 0.0	5/5	5/5	3.0	416
dSprites	1	20.0 ± 4.0	5/5	5/5	9.0	2092
dSprites	1.5	20.0 ± 4.0	5/5	5/5	9.0	2092
dSprites	2	20.0 ± 4.0	5/5	5/5	9.0	2092
dSprites	3	20.0 ± 4.0	5/5	5/5	9.0	2092
dSprites	4	20.0 ± 4.0	5/5	5/5	9.0	2092
CIFAR-10	1	24.0 ± 0.0	5/5	5/5	4.0	498
CIFAR-10	1.5	24.0 ± 0.0	5/5	5/5	4.0	498
CIFAR-10	2	24.0 ± 0.0	5/5	5/5	4.0	498
CIFAR-10	3	24.0 ± 0.0	5/5	5/5	4.0	498
CIFAR-10	4	24.0 ± 0.0	5/5	5/5	4.0	498

Table A5. Reliability-threshold sensitivity at $\beta = 1$, $c = 2$. ID: local ID start; FB: ascending fallback. Extreme tolerances are stress conditions, not recommended thresholds. Each result uses five seeds and the same reference bank. The full factorial analysis is supplied in the supplement.

$R_{\max} / A_{\max}$	Dataset	Use	Mean width	Q	Exact	Calls
0.1/0.25	MNIST	FB	6.0	5/5	5/5	4.0
0.1/0.25	Fashion-MNIST	FB	4.0	5/5	5/5	3.0
0.1/0.25	dSprites	FB	20.0	5/5	5/5	9.0
0.1/0.25	CIFAR-10	FB	24.0	5/5	5/5	4.0
0.15/0.25	MNIST	ID	6.0	5/5	5/5	9.0
0.15/0.25	Fashion-MNIST	FB	4.0	5/5	5/5	3.0
0.15/0.25	dSprites	FB	20.0	5/5	5/5	9.0
0.15/0.25	CIFAR-10	FB	24.0	5/5	5/5	4.0
0.25/0.25	MNIST	ID	6.0	5/5	5/5	9.0
0.25/0.25	Fashion-MNIST	FB	4.0	5/5	5/5	3.0
0.25/0.25	dSprites	FB	20.0	5/5	5/5	9.0
0.25/0.25	CIFAR-10	FB	24.0	5/5	5/5	4.0
0.15/0.4	MNIST	ID	6.0	5/5	5/5	9.0
0.15/0.4	Fashion-MNIST	ID	4.0	5/5	5/5	10.0
0.15/0.4	dSprites	FB	20.0	5/5	5/5	9.0
0.15/0.4	CIFAR-10	FB	24.0	5/5	5/5	4.0
0.15/0.85	MNIST	ID	6.0	5/5	5/5	9.0
0.15/0.85	Fashion-MNIST	ID	4.0	5/5	5/5	10.0
0.15/0.85	dSprites	FB	20.0	5/5	5/5	9.0
0.15/0.85	CIFAR-10	ID	24.0	5/5	5/5	4.0
0.5/2	MNIST	ID	6.0	5/5	5/5	9.0
0.5/2	Fashion-MNIST	ID	4.0	5/5	5/5	10.0
0.5/2	dSprites	ID	20.0	5/5	5/5	6.0
0.5/2	CIFAR-10	ID	24.0	5/5	5/5	4.0

Appendix B.1. Historical and Canonical Time Accounts

Table A7 preserves the MAKE50 historical ledger, including anchor completion where the original ledger omitted it. It is not the price basis of the new replay. For example, MNIST’s mean anchor prices are 69.3 s in several early ledgers and 62.6 s in the later canonical snapshot; width-six final-model prices are 39.9 and 37.6 s, respectively. The logical conditions remain `e300|es1|b1` in these matches. Identical metric summaries and requested conditions do not reveal why two timing snapshots differ, nor prove bitwise model identity. No unsupported cause such as different hardware or stopping settings is assigned.

The supplementary `results/make51/cost_lineage.csv` gives each dataset–method–weight–seed anchor/final role, complete key, original trace key, file/record source, original duration, canonical duration and available metric-identity check. Unretained off-grid candidates are marked as unavailable. An absent historical anchor charge is distinguished from a zero duration. The replay only requests retained grid candidates, so every requested anchor and final model uses a single available canonical metric/price record.

Table A6. Anchor/final timing audit at $\beta = 1$. Entries are method-role-seed records, not independent training runs. Different prices can be attached to the same logical cache key; full key, source and both durations are supplied in `cost_lineage.csv`. Stored summary metrics do not establish checkpoint identity.

Dataset	Matched entries	Different times	Max. difference (s)
MNIST	117	107	14.53
Fashion-MNIST	130	0	0.00
dSprites	119	109	0.42
CIFAR-10	130	0	0.00

Table A7. Historical MNIST training-time decomposition at $\beta = 1$ (mean seconds, five seeds). The final model is charged once and removed from the other-candidate column. ID computation and post-training evaluation were not separately timed. FONDUE epoch calibration adds 4.14 s once per dataset- β setting and is not included below.

Method	Anchor	ID AEs	Other training	Final	Total
Grid (MSE)	69.3	0.0	502.0	56.2	627.5
FONDUE	62.6	0.0	1.4	39.9	103.9
FONDUE + Q	69.3	0.0	1.4	55.3	125.9
GECO adaptation	62.6	0.0	175.4	39.6	277.6
ARD adaptation	62.6	0.0	138.0	37.6	238.2
Ascending + Q	69.3	0.0	72.8	39.9	182.0
Legacy ID window	69.3	205.5	400.7	39.9	715.4

Appendix C. Coverage and Implementation Details

590

Table A8. Executed coverage. Each image method row has five training seeds at each listed context. Additional diagnostic experiments have their own seed units.

Experiment	Dataset coverage	Contexts / counts
Fourteen ordinary selectors/controls	All four image datasets	Five β values; $14 \times 4 \times 5 \times 5 = 1400$ method records
GECO adaptation	All four	Five anchor-derived constraints; 100 records
ARD adaptation	All four	Fixed selection-stage weight one; five repeated contexts, 100 records
ID-reference bank	All four	Three widths $\times$ three seeds each, shared across contexts
Train/validation and readout trajectories	MNIST: 12, 32, 64; dSprites: 6, 16, 32	$\beta = 1$, five seeds each; 30 runs
AU prediction	Develop: MNIST; evaluate: dSprites, Fashion-MNIST, CIFAR-10	$\beta = 1$; 55 development and 155 evaluation rows
Synthetic diagnostic	Swiss roll and torus	Six capacities, three seeds, two scaling options; 72 runs

The fourteen ordinary method records comprise MSE/ELBO-proxy/probe grid selection, coarse search, fixed width, FONDUE, two FONDUE-VAR settings, ID guidance with and without AU recording, ascending+Q, and coarse, FONDUE and FONDUE-VAR with added Q. Removing AU recording does not change the ID selection decisions; its overhead was not separately timed. Machine-readable results include all fourteen, even when only the principal comparisons are printed.

591

592

593

594

595

596

Table A9. Split identities reconstructed from the saved loader and fixed seeds. Tokens are the first 16 hexadecimal characters of SHA-256 over concatenated int64 train/validation/test index arrays, matching the loader convention. Official training/test pools have separate index spaces.

Dataset	Reconstructed split token
MNIST	cffaecf5bccbb333
Fashion-MNIST	cffaecf5bccbb333
dSprites	1ea516804f7f9ac7
CIFAR-10	55e0bcf8f4474c19

The split index arrays are reconstructed from the loader specification, not recovered as an independently archived execution-time split file. The MNIST and Fashion-MNIST

597

598

tokens agree because their official pool sizes and index rules agree; their images are different. A compressed copy is supplied as `results/make51/reconstructed_split_indices.npz`. TwoNN uses Euclidean nearest-neighbour distances, excludes zero first distances and the empirical-CDF endpoint, and fits the log survival curve against the log distance ratio through the origin. The MLE cross-check averages local reciprocal mean-log-distance-ratio estimates over valid points at $k = 10$. Both estimates are floored at one in the local implementation.

Table A10. FONDUE short-training epochs and calibration time (once per dataset–context, separate from selector time). dSprites uses the source choice without calibration. A stable flag for the remaining datasets means consecutive calibration predictions agreed.

Dataset	β	Epochs	Calibration (s)	Stable
MNIST	0.25	2	9.64	yes
MNIST	0.5	2	9.55	yes
MNIST	1	1	4.14	yes
MNIST	2	2	8.12	yes
MNIST	4	2	9.38	yes
Fashion-MNIST	0.25	1	4.16	yes
Fashion-MNIST	0.5	1	3.64	yes
Fashion-MNIST	1	2	9.07	yes
Fashion-MNIST	2	3	13.70	yes
Fashion-MNIST	4	1	3.80	yes
dSprites	0.25	1	0.00	N/A (source)
dSprites	0.5	1	0.00	N/A (source)
dSprites	1	1	0.00	N/A (source)
dSprites	2	1	0.00	N/A (source)
dSprites	4	1	0.00	N/A (source)
CIFAR-10	0.25	3	30.34	yes
CIFAR-10	0.5	2	15.56	yes
CIFAR-10	1	2	22.72	yes
CIFAR-10	2	3	29.69	yes
CIFAR-10	4	4	42.98	yes

Appendix C.1. Historical Fixed-Window Procedure

The earlier procedure evaluated every candidate below its ID-derived upper width. Because its evaluated set included the quality anchor, the proposed expansion branch could not be entered. We retain this as a fixed-window audit, not an effective expanding search. Earlier V1/V2 activity checks are not selection branches; AU recording changes no selected width.

Algorithm A1 (Historical fixed-window ID procedure (audit)).

Input: splits, $\mathcal{M}$, seed s , β , 10% target margin, floor coefficient 0.001, input TwoNN d_0 , reference seeds $\{42, 123, 777\}$, thresholds $(0.15, 0.25)$, and budgets in Table 2.

1. Obtain the quality anchor and fix $T_{D,s}$ from Equation (1). In a fallback run this is an external audit step, not a constraint used by the coarse selector; its training is charged.
2. For $k \in \{4, 8, 16\}$ set $r_k = \text{round}(kd_0)$ (nearest integer; ties to even, no grid snapping). Train an FC-AE at each width for each reference seed, and compute TwoNN and MLE on the fixed 10,000 training examples. Compute R , A and $\hat{d}$.
3. If either check fails, flag ID-unreliable and evaluate the coarse subset. Its zero-based indices are the unique values of $\text{round}(\text{geomspace}(1, |\mathcal{M}|, 4)) - 1$. Return its minimum-validation-MSE candidate and continue to step 6.
4. Otherwise set $m_0 = \min\{m \in \mathcal{M} : m \geq 2\hat{d}\}$, or m_a if the set is empty. Initialise the evaluated set with the anchor. Evaluate all remaining $m \leq m_0$ in ascending order, stopping after a completed call that exceeds the budget.
5. Return the smallest target-satisfying width in the evaluated set. If it is the anchor, record no compact alternative. The anchor is always acceptable; the code's planned expansion conditional on no acceptable evaluated candidate is consequently never reached.
6. Obtain or reuse the ordinary VAE at the returned width. For $m < 1$ return a degenerate output with no model. Retain budget flags even when a model is returned. Record AU as a diagnostic without changing m .
7. Return raw and final widths, model metrics, ID diagnostics, fallback and budget flags, quality/compression indicators and the time account.

Appendix C.2. Pruning and Relevance Adaptations

The GECO adaptation uses Adam at 10^{-3} , batches of 128 in stored order, 300 epochs and hard-concrete parameters $\gamma = -0.1$, $\zeta = 1.1$ and temperature $2/3$. Initial gate log-odds are 2. The constraint is summed squared reconstruction error minus $D_{\text{input}}T_{D,s}$, with moving-average coefficient 0.99. A squared softplus multiplier is clamped to $[10^{-6}, 10^6]$ and updated by ascent on that moving-average violation. KL is gate-masked; the expected gate penalty is enabled only when the moving-average constraint is satisfied. The final readout counts deterministic gates above 0.5. Its training constraint uses squared error even on dSprites; the transferred ordinary VAE there uses Bernoulli reconstruction. Consequently the candidate likelihood convention does not imply identical pruning-stage objectives.

Table A11. Raw retained counts versus transferred ordinary-VAE widths at the primary context. Constraint denotes the GECO training moving average at the final update; Q denotes final ordinary-VAE validation quality. They evaluate different models and quantities. ARD has no GECO constraint.

Dataset	Adaptation	Raw count	Final width	Constraint	Q
MNIST	GECO	7.2 ± 0.4	6.4 ± 0.9	0/5	5/5
MNIST	ARD	6.0 ± 0.0	6.0 ± 0.0	—	5/5
Fashion-MNIST	GECO	4.2 ± 0.4	4.0 ± 0.0	1/5	5/5
Fashion-MNIST	ARD	3.8 ± 0.4	3.6 ± 0.9	—	4/5
dSprites	GECO	64.0 ± 0.0	64.0 ± 0.0	0/5	5/5
dSprites	ARD	9.2 ± 1.5	8.4 ± 1.7	—	0/5
CIFAR-10	GECO	20.0 ± 1.0	19.2 ± 4.4	5/5	2/5
CIFAR-10	ARD	14.2 ± 0.4	16.0 ± 0.0	—	0/5

For the archived GECO rows, the constraint uses sampled posterior latents and stochastic hard-concrete gates on training minibatches; it is an exponential moving average of per-example summed squared error minus the deterministic validation-anchor target multiplied by input dimension. Native validation reconstruction and trajectories were not retained. Thus 0/5 MNIST, 1/5 Fashion-MNIST, 0/5 dSprites and 5/5 CIFAR-10 endpoint constraint attainment cannot alone identify a bug or validate the implementation.

The new MNIST diagnostic retains those missing trajectories (Figure A1). Evaluation uses the full validation/test splits, posterior means or 32 posterior draws, and either deterministic continuous gate values or an explicitly binary mask. These evaluation quantities differ from stochastic training-gate reconstruction. The final training moving-average constraint is +0.045 summed-error units, with 7 counted gates and multiplier 0.844. The transferred grid width is 6. The endpoint training constraint remains violated. This diagnostic therefore does not promote the HC implementation to a validated pruning baseline.

Table A12. New MNIST HC diagnostic and dimension transfer, seed 42. $\text{MSE} \times 10^3$; target is 25.223 in the same units. HC width is the count above the gate threshold, although continuous inference retains nonbinary gate values. Sampled evaluation uses 32 latent draws and deterministic continuous gates. The ordinary VAE uses the nearest grid width.

Output / reconstruction	Width	Val. MSE	Test MSE	Q
HC, continuous/mean	7	22.424	21.950	yes
HC, continuous/sampled	7	27.019	26.519	no
HC, binary/mean	7	22.424	21.950	yes
Transferred ordinary VAE	6	22.343	21.933	yes

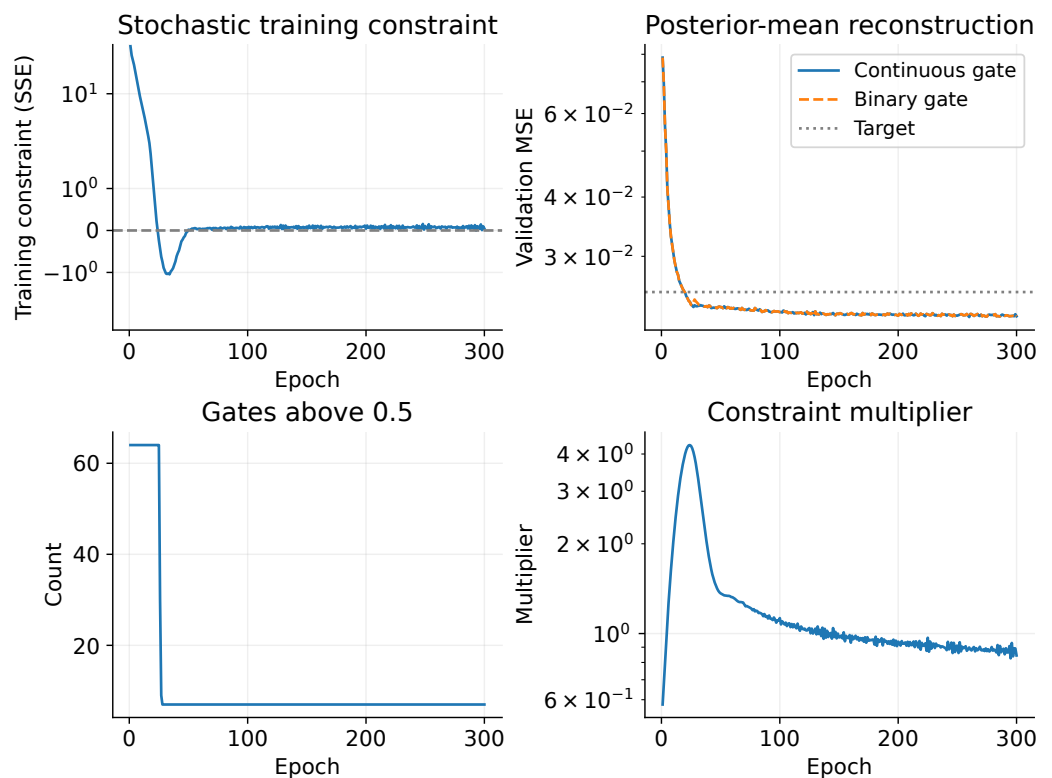


Figure A1. New MNIST hard-concrete implementation diagnostic, seed 42, 300 epochs. The training constraint uses sampled latent variables and gates; validation uses posterior means. Gate count and multiplier are shown alongside constraint and reconstruction trajectories. This is one local HC run, not an original ARM reproduction.

The ARD adaptation reserves the first 10% (1800 examples) of its training array for prior estimation and trains on the remainder. It updates $p_j = \text{mean}(\mu_j^2 + \sigma_j^2)$ every ten epochs and after training, clipping to $[10^{-6}, 10^6]$. This uses analytic posterior second moments in a fixed subset, rather than a periodically resampled encoded-data pool. Selection

training uses reconstruction plus Gaussian KL at unit weight. On the first 512 training examples it computes

$$w_j = \text{mean}_{x,k} [g_k(\mu(x) + \sqrt{p_j} e_j) - g_k(\mu(x))]^2, \quad q_j = w_j p_j, \quad (\text{A1})$$

and keeps the smallest number of descending scores reaching 99% of their total. The finite difference is not divided by its squared step. Locally it already contains a factor p_j , so multiplication by p_j again differs from a Jacobian-squared times variance score. The stored counts are therefore labelled an *ARD adaptation*; this difference cannot be removed by correcting the prose. The Gaussian approximation alone is not an original-source deviation. Identical relevance arrays and counts across contexts are checked in the numerical audit; they represent repeated fixed settings, not a β sweep.

The stored q_j and p_j are sufficient to undo the extra finite-step factor without access to the checkpoint. With $h_j = \sqrt{p_j}$, the correctly normalized finite-step score is

$$\tilde{q}_j = p_j \text{mean}_{x,k} \left[\frac{g_k(\mu + h_j e_j) - g_k(\mu)}{h_j} \right]^2 = \frac{q_j}{p_j}. \quad (\text{A2})$$

All saved p_j are positive. We recompute the cumulative 99% count and transfer it using the same nearest-grid rule. Table A13 shows that CIFAR-10 changes from a mean raw count 14.2 to 26.0, mapping to width 24 and attaining Q in all five trials. This is an algebraic correction of the finite-step readout up to stored numerical precision. Its possibly large steps and altered prior updates remain; it is not an exact-Jacobian or original-ARD benchmark.

Table A13. Corrected finite-step relevance readout from saved scores and prior variances at the primary context. The same stored perturbations are normalized by their squared step before variance weighting; no new ARD training is involved. This validates an algebraic readout correction, not exact Jacobian relevance or the source training procedure. Test MSE $\times 10^3$ is for transferred ordinary VAEs.

Dataset	Old raw	Corrected raw	Grid width	Q	Test MSE
MNIST	6.0 ± 0.0	6.2 ± 0.4	6.0 ± 0.0	5/5	22.05 ± 0.17
Fashion-MNIST	3.8 ± 0.4	4.0 ± 0.0	4.0 ± 0.0	5/5	19.06 ± 0.25
dSprites	9.2 ± 1.5	10.6 ± 0.5	10.0 ± 0.0	0/5	1.47 ± 0.05
CIFAR-10	14.2 ± 0.4	26.0 ± 0.0	24.0 ± 0.0	5/5	15.36 ± 0.13

The FONDUE description-based classifier was checked against the public repository https://github.com/bonheml/VAE_learning_dynamics at commit f6d6e66b536bcbe635392d3438338949d4dc8730. The local implementation uses the stated posterior-variance categories with one exponentiation and TwoNN; equivalence to the public execution path and the original experiment configuration is not asserted. As a historical reference, the source reports dSprites selection 12.2 ± 0.4 at one epoch [4], versus 4.2 ± 0.4 here at $\beta = 1$. This unmatched-setting difference is not proof of an implementation error, nor a reproduction of the original benchmark. A separate control-flow check replays the source algorithm's lower/upper-bound updates on the archived ID-gap sequence: all 100 FONDUE paths and returned widths agree with that pseudocode and terminate normally. This verifies the search update logic on those inputs; it does not validate the estimator implementation, training framework parity or the published benchmark. Its row-level results are included as `results/make51/fondue_control_flow.json`.

Appendix C.3. Threshold-Only and Quality-Margin Analyses

The retrospective predictive analysis uses Python 3.11.11, NumPy 2.3.2 and scikit-learn 1.7.1. Table A14 documents probability saturation in this reconstruction. The original

predictor’s complete environment and row-level probabilities were not saved, so its exact floating-point results cannot be verified from the summary alone. Row-level decision scores and probabilities for the current reconstruction are saved in `results/make51/au_predictions.csv`.

Table A14. Numerical probability saturation in the current predictor reconstruction. Entries count predictions rounded to exactly one, before AUROC calculation. Decision scores preserve ordering among these rows.

Evaluation condition	Base	Base + AU	Base + random
dSprites	55/55	44/55	49/55
Fashion-MNIST	0/55	0/55	0/55
CIFAR-10	45/45	40/45	36/45

The main AU-threshold figure uses identical checkpoints for all three δ values. Across the full ladder, the fractions of CIFAR-10 grid/seed candidates with AU equal to capacity are given below. These counts use ten widths and five seeds per β ; their rows are not 50 independent datasets.

Table A15. CIFAR-10 (Gaussian): percentage of the 50 width/seed candidates with AU equal to capacity. Thresholds are re-read on the same checkpoints within each column.

δ	$\beta = .25$	.5	1	2	4
0.001	100	100	100	98	60
0.01	60	50	36	22	10
0.1	48	38	32	20	10

For a separate retrospective quality-margin check, the complete cached grid at $\beta = 1$ is held fixed and the smallest satisfying width is read out for each margin, without retraining or changing the anchor. Table A16 reports this full-grid criterion, not the path or cost of any restricted-window selector. No FONDUE-threshold or ARD-99% sensitivity training was executed, so no response of those rules to their own readout thresholds is claimed.

Table A16. Smallest quality-satisfying cached width at $\beta = 1$ as relative margin changes; five seed means $\pm$ SD. The absolute floor 0.001 times validation variance remains fixed. Gaussian likelihood except Bernoulli dSprites.

Dataset	1%	5%	10%	25%	50%
MNIST	6.0 $\pm$ 0.0	6.0 $\pm$ 0.0	6.0 $\pm$ 0.0	4.0 $\pm$ 0.0	4.0 $\pm$ 0.0
Fashion-MNIST	4.0 $\pm$ 0.0	4.0 $\pm$ 0.0	4.0 $\pm$ 0.0	4.0 $\pm$ 0.0	2.0 $\pm$ 0.0
dSprites	51.2 $\pm$ 17.5	36.0 $\pm$ 16.5	20.0 $\pm$ 4.0	12.0 $\pm$ 0.0	9.6 $\pm$ 0.9
CIFAR-10	41.6 $\pm$ 8.8	32.0 $\pm$ 0.0	24.0 $\pm$ 0.0	16.0 $\pm$ 0.0	16.0 $\pm$ 0.0

Appendix D. Synthetic Scale and Ablation Diagnostic

Swiss-roll points are $(t \cos t, h, t \sin t)$ with $t \sim U[1.5\pi, 4.5\pi]$ and $h \sim U[0, 1]$. Torus points are $((3 + \cos \phi) \cos \theta, (3 + \cos \phi) \sin \theta, \sin \phi)$ with independent angles uniform on $[0, 2\pi]$. Each has 5000 examples, generation seed zero, and independent Gaussian coordinate noise with SD 0.01. The first 4500 are used for fitting and the remaining 500 for the reported diagnostic. Per-coordinate standardisation in the stored experiment was fitted to all 5000 before splitting. This is disclosed as preprocessing leakage; the results are descriptive, not clean held-out test estimates.

The FC VAE uses widths 128 and 64, reversed in its decoder, ReLU hidden activations, a linear output, Adam at 10^{-3} , batch 128, 300 epochs, $\beta = 1$ and seeds 42, 123 and 777. At the last checkpoint, coordinates with posterior-mean variance at most 0.01 on the 500

diagnostic examples are zeroed and MSE is recomputed on those same examples. Table A17 reports the standardized runs. There is no separate test set in this stored experiment.

Table A17. Synthetic scale/ablation diagnostic, Gaussian likelihood, $\beta = 1$, $\delta = 0.01$, three training seeds. Data were standardized before the split; these are descriptive validation results, not leakage-free generalization estimates. Changes are mean per-run relative MSE changes (%).

Data	m	AU	MSE before	MSE after	Change
Swiss roll	1	1.0 $\pm$ 0.0	0.3511 $\pm$ 0.0267	0.3511 $\pm$ 0.0267	+0.000
Swiss roll	2	2.0 $\pm$ 0.0	0.1997 $\pm$ 0.0177	0.1997 $\pm$ 0.0177	+0.000
Swiss roll	3	3.0 $\pm$ 0.0	0.1407 $\pm$ 0.0018	0.1407 $\pm$ 0.0018	+0.000
Swiss roll	4	3.0 $\pm$ 0.0	0.1404 $\pm$ 0.0136	0.1406 $\pm$ 0.0135	+0.147
Swiss roll	6	3.0 $\pm$ 0.0	0.1365 $\pm$ 0.0023	0.1368 $\pm$ 0.0022	+0.183
Swiss roll	8	3.0 $\pm$ 0.0	0.1437 $\pm$ 0.0089	0.1440 $\pm$ 0.0089	+0.165
Torus	1	1.0 $\pm$ 0.0	0.3217 $\pm$ 0.0163	0.3217 $\pm$ 0.0163	+0.000
Torus	2	2.0 $\pm$ 0.0	0.1654 $\pm$ 0.0242	0.1654 $\pm$ 0.0242	+0.000
Torus	3	3.0 $\pm$ 0.0	0.1083 $\pm$ 0.0036	0.1083 $\pm$ 0.0036	+0.000
Torus	4	3.0 $\pm$ 0.0	0.1124 $\pm$ 0.0050	0.1124 $\pm$ 0.0050	+0.063
Torus	6	3.0 $\pm$ 0.0	0.1095 $\pm$ 0.0035	0.1096 $\pm$ 0.0035	+0.107
Torus	8	3.0 $\pm$ 0.0	0.1104 $\pm$ 0.0069	0.1105 $\pm$ 0.0069	+0.156

Both noise-free generating surfaces have intrinsic dimension two and are represented here in three ambient coordinates. A Swiss-roll patch does not thereby have minimum embedding dimension three. Observing three active coordinates under this loss is not a topological theorem. The earlier manuscript said its synthetic coordinates were standardized; the implementation audit found a mismatch between that description and the executed preprocessing. On the unstandardized Swiss roll the height coordinate has much smaller variance than the other coordinates, making its omission inexpensive in squared error. This scale observation does not alone prove the complete causal explanation of every earlier AU result.

Appendix E. New MNIST Model Records

The new experiment uses all twelve declared widths, training seed 42, and the original split reconstructed from its fixed seeds. The model architecture and objective match the ordinary MNIST protocol, while the CPU environment differs from the archived CUDA runs. Table A18 gives every proxy-selected checkpoint’s validation criteria. MC SE refers only to evaluation noise over posterior draws. The frozen capacity-choice file precedes final test evaluation. Weights, epoch curves and draw-wise ELBO means are included in the reviewer supplement. No between-training-seed confidence interval is inferred from this one-seed validation.

Table A18. Complete newly trained MNIST grid, seed 42, $\beta = 1$, Gaussian likelihood. Epoch selected by the deterministic proxy; stop is the actual final training epoch. Proxy omits the Gaussian constant and posterior expectation; MC ELBO includes both. MSE $\times 10^3$.

Width	Epoch	Stop	Val. MSE	Proxy	MC ELBO $\pm$ SE
2	81	111	37.406	-34.647	-484.386 $\pm$ 0.009
4	85	115	27.840	-29.304	-480.108 $\pm$ 0.009
6	169	199	22.343	-26.696	-478.174 $\pm$ 0.010
8	73	103	22.417	-26.544	-478.285 $\pm$ 0.009
10	70	100	22.954	-26.663	-478.627 $\pm$ 0.009
12	196	226	20.983	-26.055	-478.067 $\pm$ 0.012
16	107	137	23.030	-26.774	-478.592 $\pm$ 0.009
20	144	174	21.327	-26.156	-478.134 $\pm$ 0.010
24	132	162	22.977	-26.956	-478.664 $\pm$ 0.008
32	136	166	23.402	-27.042	-478.869 $\pm$ 0.012
48	114	144	24.177	-27.411	-479.310 $\pm$ 0.013
64	217	247	22.930	-26.946	-479.043 $\pm$ 0.011

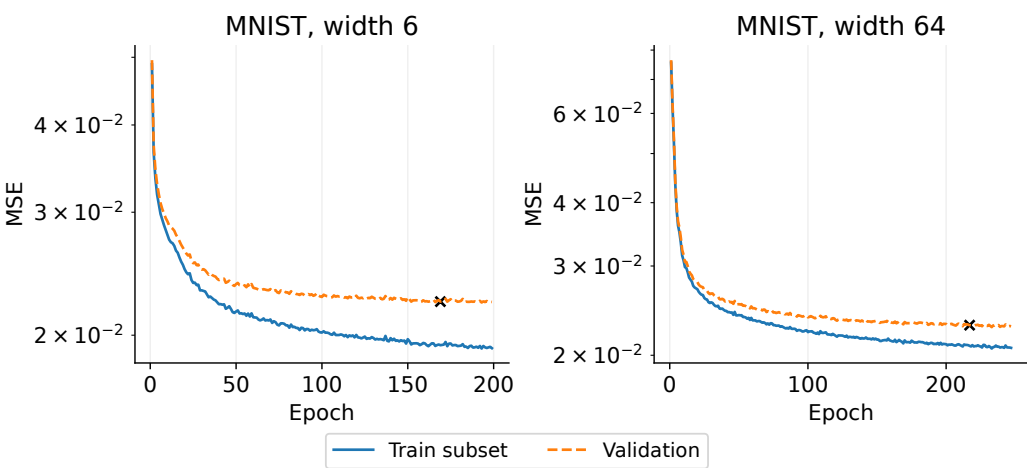


Figure A2. New MNIST seed-42 train/validation MSE at widths six and 64, Gaussian likelihood and $\beta = 1$. Training curves use a fixed 2000-example subset; validation uses all 2000 held-out examples. Black crosses mark proxy-selected epochs. These are new saved runs, not reconstructed train curves for the historical CUDA checkpoints.

Appendix F. Reproducibility Inventory and Remaining Records

The reaggregation entry point is `src/make51/rebuild_review.py`; it reads saved JSON and configuration files and writes numerical tables, six figures and the paired predictive analysis from the historical records. The new search analysis is `src/make51/replay_search.py`; the bounded model validation is `src/make51/validate_models.py`. Its outputs are summarized by `src/make51/summarize_validation.py`. The manuscript assembly entry point is `src/make51/build_manuscript.py`. The generated main TeX embeds its table contents and references the figure PDFs. Compile from the project root with `pdflatex` repeatedly until references settle. The accompanying `MAKE51_reproducibility.zip` supplies the files, not only this inventory. Its README distinguishes inexpensive reaggregation from optional retraining, gives dependency versions and lists unavailable historical artifacts. `MANIFEST.sha256` identifies the included payload. The archive contains a fixed source snapshot and is usable without access to a private repository.

Table A19. Source-to-output map for the present revision. Paths below are relative to the project root.

Evidence	File or directory
Configuration	<code>config/experiment_config_MAKE49.json</code>
Ordinary method records	<code>results/make49/stage4_methods.json</code>
Pruning records	<code>results/make49/stage7_b7b8.json</code>
Candidate metrics and latent variances	<code>results/make49/stage4_cache/</code>
Train/validation and readout trajectories	<code>results/make49/stage3_e7a_e2.json</code>
ID representation diagnostics	<code>results/make49/stage5_e3_reference.json</code>
Original AU prediction result	<code>results/make49/stage5_e4_au_value.json</code>
Synthetic runs	<code>results/make49/stage6_e6_synthetic.json</code>
All audited method rows	<code>results/make51/method_results.csv</code>
Paired AU analysis	<code>results/make51/au_paired.json</code>
Search replay and sensitivity	<code>results/make51/search_replay.csv</code> ; <code>results/make51/search_sensitivity.csv</code>
Timing lineage	<code>results/make51/cost_lineage.csv</code>
New model records	<code>results/make51/model_validation/</code>
Snapshot identities	<code>results/make51/source_manifest.json</code>
Figures	<code>results/figures/MAKE51/</code>

Method-level row identifiers are (dataset, method, candidate seed, β or context). Candidate keys also contain width, requested epochs and stopping option. Every configuration/dataset–method–context cell in the main method archive has five seeds;

this is checked during reaggregation. The ARD repetition does not create five distinct selection-weight conditions. Individual early off-grid cache entries are not all retained, although their final metrics and training traces remain in the method records. The original test predictions, historical full model checkpoints and independent ID/evaluation timings are not reconstructed from missing files.

The local change record documents the initial specification on 12 September; the switch to Bernoulli dSprites after collapse diagnostics; the move of ID estimation from an impossible 10,000-example validation sample to training data; trajectory runs without early stopping; and the 13 September reference-width, run-accounting and degenerate-output clarifications. The random-feature control was added after the primary predictive analysis. In this revision, anchor-cost completion, paired AU inference, quality-margin rereading and explicit implementation auditing are retrospective. None is relabelled as a publicly preregistered test. The 19 September local policies, threshold factorial, normalized ARD readout and new one-seed validation were specified after those earlier results. The supplement preserves their distinct specifications and output locations.

References

- Kingma, D.P.; Welling, M. Auto-encoding variational Bayes. In *Proceedings of the 2nd International Conference on Learning Representations (ICLR)*, Banff, AB, Canada, 14–16 April 2014. arXiv:1312.6114. <https://doi.org/10.48550/arXiv.1312.6114>.
- Facco, E.; d’Errico, M.; Rodriguez, A.; Laio, A. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Sci. Rep.* **2017**, *7*, 12140. <https://doi.org/10.1038/s41598-017-11873-y>.
- Levina, E.; Bickel, P.J. Maximum likelihood estimation of intrinsic dimension. In *Advances in Neural Information Processing Systems*, **2004**, *17*. <https://proceedings.neurips.cc/paper/2004/hash/74934548253bcab8490ebd74afed7031-Abstract.html>.
- Bonheme, L.; Grzes, M. FONDUE: An algorithm to find the optimal dimensionality of the latent representations of variational autoencoders. arXiv **2022**, arXiv:2209.12806. <https://doi.org/10.48550/arXiv.2209.12806>.
- Burda, Y.; Grosse, R.; Salakhutdinov, R. Importance weighted autoencoders. In *Proceedings of the 4th International Conference on Learning Representations (ICLR)*, San Juan, Puerto Rico, 2–4 May 2016. arXiv:1509.00519. <https://doi.org/10.48550/arXiv.1509.00519>.
- Bonheme, L.; Grzes, M. Be more active! Understanding the differences between mean and sampled representations of variational autoencoders. *J. Mach. Learn. Res.* **2023**, *24*(324), 1–30. <https://jmlr.org/papers/v24/21-1145.html>.
- De Boom, C.; Wauthier, S.; Verbelen, T.; Dhoedt, B. Dynamic narrowing of VAE bottlenecks using GECCO and L_0 regularization. arXiv **2020**, arXiv:2003.10901. <https://doi.org/10.48550/arXiv.2003.10901>.
- Saha, S.; Joshi, S.; Whitaker, R. ARD-VAE: A statistical formulation to find the relevant latent dimensions of variational autoencoders. arXiv **2025**, arXiv:2501.10901. <https://doi.org/10.48550/arXiv.2501.10901>.
- Tipping, M.E.; Bishop, C.M. Probabilistic principal component analysis. *J. R. Stat. Soc. Ser. B* **1999**, *61*, 611–622. <https://doi.org/10.1111/1467-9868.00196>.
- Tenenbaum, J.B.; de Silva, V.; Langford, J.C. A global geometric framework for nonlinear dimensionality reduction. *Science* **2000**, *290*, 2319–2323. <https://doi.org/10.1126/science.290.5500.2319>.
- Roweis, S.T.; Saul, L.K. Nonlinear dimensionality reduction by locally linear embedding. *Science* **2000**, *290*, 2323–2326. <https://doi.org/10.1126/science.290.5500.2323>.
- Ceruti, C.; Bassis, S.; Rozza, A.; Lombardi, G.; Casiraghi, E.; Campadelli, P. DANCo: An intrinsic dimensionality estimator exploiting angle and norm concentration. *Pattern Recognit.* **2014**, *47*, 2569–2581. <https://doi.org/10.1016/j.patcog.2014.02.013>.

13. Johnsson, K.; Sonesson, C.; Fontes, M. Low bias local intrinsic dimension estimation from expected simplex skewness. *IEEE Trans. Pattern Anal. Mach. Intell.* **2015**, *37*, 196–202. <https://doi.org/10.1109/TPAMI.2014.2343220>.
14. Costa, J.A.; Hero, A.O. Geodesic entropic graphs for dimension and entropy estimation in manifold learning. *IEEE Trans. Signal Process.* **2004**, *52*, 2210–2221. <https://doi.org/10.1109/TSP.2004.831130>.
15. Camastra, F.; Staiano, A. Intrinsic dimension estimation: Advances and open problems. *Inf. Sci.* **2016**, *328*, 26–41. <https://doi.org/10.1016/j.ins.2015.08.029>.
16. Allegra, M.; Facco, E.; Denti, F.; Laio, A.; Mira, A. Data segmentation based on the local intrinsic dimension. *Sci. Rep.* **2020**, *10*, 16449. <https://doi.org/10.1038/s41598-020-72222-0>.
17. Gałka, Ł.; Karczmarek, P.; Tokovarov, M. Isolation Forest Based on Minimal Spanning Tree. *IEEE Access* **2022**, *10*, 74175–74186. <https://doi.org/10.1109/ACCESS.2022.3190505>.
18. Ansuini, A.; Laio, A.; Macke, J.H.; Zoccolan, D. Intrinsic dimension of data representations in deep neural networks. In *NeurIPS 2019*, Vol. 32, 2019. https://proceedings.neurips.cc/paper_files/paper/2019/file/cfccc0621b49c983991ead4c3d4d3b6b-Paper.pdf.
19. Pope, P.; Zhu, C.; Abdelkader, A.; Goldblum, M.; Goldstein, T. The intrinsic dimension of images and its impact on learning. In *ICLR 2021*, 2021. arXiv:2104.08894. <https://doi.org/10.48550/arXiv.2104.08894>.
20. Rezende, D.J.; Viola, F. Taming VAEs. arXiv **2018**, arXiv:1810.00597. <https://doi.org/10.48550/arXiv.1810.00597>.
21. Louizos, C.; Welling, M.; Kingma, D.P. Learning sparse neural networks through L_0 regularization. In *Proceedings of the 6th International Conference on Learning Representations (ICLR)*, Vancouver, BC, Canada, 30 April–3 May 2018. arXiv:1712.01312. <https://doi.org/10.48550/arXiv.1712.01312>.
22. Obata, C.; Jin'no, K. Analysis of the relationship between manifold structure and latent space capacity of input data using VAE. In *Proceedings of the 2025 International Symposium on Nonlinear Theory and Its Applications (NOLTA2025)*, Okinawa, Japan, 27–31 October 2025; pp. 32–35. Publication record: https://www.comm.tcu.ac.jp/nel/list_j_2025.html (accessed on 18 September 2026).
23. Whitney, H. The self-intersections of a smooth n -manifold in $2n$ -space. *Ann. Math.* **1944**, *45*, 220–246. <https://doi.org/10.2307/1969265>.
24. Lucas, J.; Tucker, G.; Grosse, R.B.; Norouzi, M. Don't blame the ELBO! A linear VAE perspective on posterior collapse. In *NeurIPS 2019*, Vol. 32, 2019. <https://proceedings.neurips.cc/paper/2019/file/7e3315fe390974fcf25e44a9445bd821-Paper.pdf>.
25. Rolinek, M.; Zietlow, D.; Martius, G. Variational autoencoders pursue PCA directions (by accident). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA, 15–20 June 2019; pp. 12406–12415. <https://doi.org/10.1109/CVPR.2019.01269>.
26. LeCun, Y.; Bottou, L.; Bengio, Y.; Haffner, P. Gradient-based learning applied to document recognition. *Proc. IEEE* **1998**, *86*, 2278–2324. <https://doi.org/10.1109/5.726791>.
27. Xiao, H.; Rasul, K.; Vollgraf, R. Fashion-MNIST: A Novel Image Dataset for Benchmarking Machine Learning Algorithms. *arXiv* **2017**, 1708.07747. <https://doi.org/10.48550/arXiv.1708.07747>.
28. Krizhevsky, A. Learning multiple layers of features from tiny images. *Technical Report*; University of Toronto: Toronto, ON, Canada, 2009.
29. Matthey, L.; Higgins, I.; Hassabis, D.; Lerchner, A. dSprites: Disentanglement testing sprites dataset. 2017. Available online: <https://github.com/google-deepmind/dsprites-dataset> (accessed on 12 September 2026).
30. Fu, H.; Li, C.; Liu, X.; Gao, J.; Celikyilmaz, A.; Carin, L. Cyclical annealing schedule: A simple approach to mitigating KL vanishing. In *Proceedings of NAACL-HLT 2019*, Minneapolis, MN, USA, 2019; pp. 240–250. <https://doi.org/10.18653/v1/N19-1021>.